IneQuality*

Gideon Bornstein†
Alessandra Peter‡

August 20, 2026

Abstract

This paper studies the distributional consequences of market power when firms endogenously design their product portfolios. We develop a general equilibrium model in which oligopolistic firms choose product quality and prices while competing both with rivals and across their own products. Firms with substantial market power degrade low-end products and raise their prices to limit cannibalization of high-end sales; firms facing sufficient competition do not. These distortions disproportionately harm low-income households, while high-income households may benefit from imperfect competition through general equilibrium forces. We provide empirical evidence supporting the mechanism and calibrate a quantitative multi-sector model to the U.S. economy. We find that market power lowers low-income households' welfare by over 7 log points and raises high-income households' welfare by 1.6 log points relative to the perfectly-competitive allocation. The calibrated model suggests that the rise in concentration since the early 1990s widened the welfare gap between high- and low-income households by 2.1 log points.

*We thank Pete Klenow, Dirk Krueger, Virgiliu Midrigan, Simon Mongey, Ezra Oberfield, and Galina Vereshchagina (discussant) for valuable comments and suggestions, as well as seminar participants at the SED, Normac, AEA, Wharton, NYU, Boston University, Boston College, Notre Dame, Cornell, Yale, the Richmond Fed Core Week, NBER Summer Institute, and the Minnesota Workshop in Macroeconomic Theory for helpful feedback. Jacky Chen provided excellent research assistance. Researchers' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the researchers and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

†The Wharton School, University of Pennsylvania and NBER; gideonbo@wharton.upenn.edu
‡Department of Economics, New York University, NBER, and CEPR; alessandra.peter@nyu.edu

1 Introduction

Over the past four decades, U.S. product markets have grown substantially more concentrated. The employment share of the largest firms has increased across most sectors (Autor et al., 2020) and so have markups (De Loecker et al., 2020). A large literature has studied the aggregate efficiency consequences of this rise in market power. But market power may also have *distributional* consequences. Rich and poor households often purchase different products from the same firm: a Toyota Camry or a Lexus, a coach seat or business class, an iPhone SE or an iPhone Pro Max. How firms design these product portfolios, and the prices at which they offer them, depends on the competitive pressure they face from rival firms. This paper studies how market power shapes the product design choices of firms and, through this channel, affects consumption inequality.

To study the distributional consequences of market power, we proceed in three steps. First, we develop a general equilibrium model in which oligopolistic firms endogenously choose the quality and prices of their products. Firms compete both externally with other firms in the sector and internally through cannibalization across their own product lines. We show that only high-market-power firms have cannibalization concerns: these firms distort downward the quality of their low-end goods, raise their low-end prices, and lower the prices of their high-end products. By contrast, low-market-power firms, which face sufficient external competition, have no incentive to distort their product design. As a result, an increase in market power hurts low-income households disproportionately, while general equilibrium forces can even make high-income households better off in absolute terms. Second, we provide empirical evidence for the model's key prediction using data on grocery store goods, showing that in more concentrated product markets, the largest firms sell disproportionately more to high-income consumers. Third, we calibrate the model to quantify the distributional effects of market power. We find that, relative to the perfectly-competitive allocation, market power lowers the welfare of low-income households by 7.3 log points while raising that of high-income households by 1.6 log points.

The key mechanism behind the distributional impact of market power is *cannibalization-driven quality distortion*. When a firm with market power sells products to consumers of different income levels, it faces an internal trade-off: making the low-end product too attractive risks cannibalizing sales of its high-end offering. To prevent high-income consumers from downgrading, the firm deliberately reduces the quality of its low-end product below the efficient level and raises its price. This mechanism is muted with sufficient competition, as firms that degrade their products would simply lose customers to competitors. Our model extends Mussa and Rosen (1978), who study endogenous quality choices of a monopolist, to an environment with oligopolistic competition in general equilibrium.

The model we construct features a continuum of sectors, each populated by a finite number of firms that differ in their brand value. Consumers differ in income and, within each sector, choose a single product from a single firm. Each firm designs a menu of vertically differentiated products from which consumers self-select according to their income. There are two sources of market power in our model: the number of firms in a sector and the dispersion in brand values across firms. The model's structure allows us to isolate each source and to decompose the welfare costs of market power into those that fall symmetrically on all households and those that disproportionately burden the poor.

We develop our argument in two steps. We start by analyzing an "island economy" in which high- and low-income consumers are segregated into separate markets, so that cannibalization concerns are absent. In this benchmark, imperfect competition still generates misallocation across firms: high-brand-value firms charge higher markups and attract too few consumers, a distortion in the spirit of Atkeson and Burstein (2008). However, product design remains efficient. All firms offer quality levels that equate the marginal utility of quality to its marginal cost of production, just as the social planner would. As a result, consumption inequality equals income inequality, regardless of the degree of market power. The island economy isolates the standard welfare cost of oligopolistic competition: prices that are too high and too little consumption from the largest firms. This welfare cost, while present, falls equally on rich and poor households.

In the full market equilibrium, high- and low-income consumers are no longer segregated, and the island allocation may not be sustainable. High-brand-value firms charge higher markups in the island economy, so the price gap between their high- and low-end products is large. If the markup is large enough, high-income consumers at these firms would prefer to purchase the cheaper low-end product and use the savings in other sectors. To prevent this, firms with sufficient market power distort the quality of their low-end product downward and raise its price, while reducing the price of their high-end product. In general equilibrium, the resources freed by the reduction in low-end quality must be absorbed elsewhere. Some of these resources are redirected toward the goods consumed by high-income households, whose equilibrium quality rises above the efficient level. Through this general equilibrium channel, high-income households benefit from the very distortions that hurt the poor.

There are two sources of market power in the model: the number of competing firms and the dispersion in brand value across firms. Both sources increase consumption inequality. When the number of firms falls, each firm's market share rises, leading to an increase in desired markups. With higher markups, there is more scope for cannibalization and firms distort quality at the bottom more. When brand-value dispersion increases, "superstar" firms emerge with disproportionately large market shares. These firms face the strongest cannibalization incentives, so they distort their low-end products the most and reduce prices of high-end products. As a result, misallocation of consumers across firms worsens for low-income consumers, who are pushed toward lower-brand-value firms, and improves for high-income ones, who gravitate toward the now-cheaper high-end products.

The model delivers a sharp empirical prediction: in more concentrated markets, dominant firms should have a higher market share among high-income households relative to low-income ones. We test this prediction using the NielsenIQ Consumer Panel, which records household-level purchases of grocery products along with household demographics. For each product market, we construct the relative market share of the top firm across income groups. We find that in more concentrated product markets, the top firm's relative market share tilts more toward the high-income segment. In the typical market, the top firm's market share among high-income households is about 4 log points higher than among low-income ones. A one standard deviation increase in concentration, as measured by the Herfindahl-Hirschman Index (HHI), more than doubles this gap. This relationship holds when we use alternative measures of concentration, or when we broaden the definition of top firms to the largest three.

To quantify the distributional effects of market power, we calibrate the model to the 2022 U.S. economy, targeting spending inequality from the PSID, the number of firms per 6-digit NAICS industry, and the top-4 sales share across these industries. In the calibrated model, market power reduces the welfare of low-income households by 7.3 log points relative to the perfectly-competitive allocation, while high-income households are 1.6 log points better off. To isolate the role of product design, we compare these results to the island economy, which features the same degree of imperfect competition but shuts down cannibalization across income groups. There, the welfare loss is a symmetric 1.4 log points for both income groups. The distributional costs of market power are several times larger than the standard efficiency costs. They are also large enough to flip the sign for high-income households: the gains from lower high-end prices and higher quality more than offset the standard welfare costs of imperfect competition.

We then use the model to assess the distributional impact of the rise in market concentration since the early 1990s. We model this rise as an increase in the dispersion of brand values across firms, consistent with the emergence of dominant superstar firms. The increase in concentration over this period widened the welfare gap between high- and low-income households by 2.1 log points. Our analysis shows that less competition, even if driven by the emergence of superstar firms and not merely through a reduction in the number of competitors, leads to more product distortion and, through it, higher consumption inequality.

Related literature. This paper connects to several strands of work.

First, it contributes to the literature on the aggregate consequences of rising market power. A large literature has studied the macroeconomic implications of market power, focusing on efficiency costs and distributional consequences operating through aggregate factor income shares, e.g., (Hsieh and Klenow, 2009; De Loecker et al., 2020; Baqaee and Farhi, 2020; Autor et al., 2020; Edmond et al., 2022; Boar and Midrigan, 2024). Our paper shows that market power also has distributional consequences that operate through a distinct channel: the endogenous design of product portfolios.

A large literature has studied the macroeconomic implications of market power, focusing on efficiency costs and distributional consequences operating through aggregate factor income shares

Second, our paper is related to the theoretical literature on product quality distortions. Starting with Mussa and Rosen (1978), this literature shows that a monopolist may distort the quality offered to low-valuation consumers. Unlike that canonical framework, firms in our setting compete for customers with other firms in the sector. Two related papers that consider oligopolistic competition are Stole (1995) and Rochet and Stole (2002), both of which show that intensive competition reduces the scope of product quality distortions. Our analysis adds to these papers by developing a tractable general equilibrium framework which allows us to study how market structure shapes the distribution of consumption across income groups and to quantify aggregate welfare effects. We show that, unlike in the partial-equilibrium comparative statics, general equilibrium forces can imply that high-income households benefit from a lack of competition.

Finally, the paper is related to a literature studying how prices, markups, and product availability vary with household income. Jaravel (2019) and Oberfield (2023) study whether product innovations disproportionally benefit high- or low-income consumers. Kaplan and Menzio (2015), Nord (2023), and Sangani (2024) document large price dispersion across stores and relate it to household search behavior. Becker (2024) and Mongey and Waugh (2025) study the role of non-homothetic preferences in firm pricing. Relative to this literature, we endogenize the set of products that are offered to consumers and find that distortions in quality hurt low-income consumers more than markups.

Layout. The paper proceeds as follows. Section 2 presents the model, defines equilibrium, and characterizes the perfectly competitive allocation. Section 3 analyzes the island economy and the market equilibrium, and derives comparative statics linking market power to consumption inequality. Section 4 presents empirical evidence. Section 5 describes the calibration and quantitative results. Section 6 concludes.

2 Model

This section develops a general–equilibrium model in which firms endogenously choose the quality levels of the goods they sell. Households differ in their income levels and purchase a single good in each sector. We first describe preferences and household demand, then turn to firm behavior, and finally define the competitive equilibrium.

2.1 Environment

Households. The economy is populated by two types of households, low- and high-income ones. There is a measure one-half of each type of household. Labor supply is inelastic. High-income households have high labor endowment $L_h = 1 + \alpha$ and receive a higher share of aggregate profits $(1 + \alpha)$. Low-income households have low labor endowment $L_l = 1 - \alpha$ and receive the remainder $(1 - \alpha)$ of aggregate profits. We denote households by $i \in (0, 1)$, where households in $(0, \frac{1}{2})$ are low-income and households in $(\frac{1}{2}, 1)$ are high-income.

Households consume products from a continuum of sectors, indexed by $s \in (0, 1)$. The aggregate consumption is a Cobb-Douglas aggregator across sectoral consumption,

$$\ln C_i = \int_0^1 \ln c_i(s) ds, \tag{2.1}$$

where $c_i(s)$ is the utility-adjusted consumption of household i from sector s. Within each sector, there is a finite number of firms, N, each offering a menu of vertically differentiated products. Products differ in the quality level q_{jk}, where jk is the k-th product sold by firm j. Households face a discrete choice problem within each sector, i.e., they choose to purchase a single product k from a single firm j.

Utility-adjusted consumption consists of three components. First, households derive utility from a product's quality. Second, households have preferences towards the products sold by firm j. This captures both brand value and non-rival product features, such as access to the iOS operating system. We assume that these come in the form of taste shifters and have a common component η_j and an idiosyncratic component $\varepsilon_{ij}/(\sigma-1)$.[1] The idiosyncratic taste shifter ε_{ij} is independent across households and sectors and follows a standard Gumbel distribution. This assumption allows us to characterize the demand function of households analytically. Formally, the utility-adjusted consumption from product jk is given by

$$\ln c_{ijk} = \ln q_{jk} + \eta_j + \frac{\varepsilon_{ij}}{\sigma - 1}. \tag{2.2}$$

Households solve a collection of discrete choice problems: In each sector s, they choose the firm $j(s)$ and product $k(s)$ to maximize their utility, subject to their type's budget constraint.

$$\begin{aligned} \max_{\{j(s),k(s),c_i(s)\}} \quad & \int \ln c_i(s) ds \\ \text{s.t.} \quad & \int p_{j(s)k(s)} ds = E_i, \\ & \ln c_i(s) = \ln q_{j(s)k(s)} + \eta_{j(s)} + \frac{\varepsilon_{ij(s)}}{\sigma - 1} \end{aligned} \tag{2.3}$$

Let P_i denote the inverse Lagrange multiplier of the problem above. That is, P_i measures the cost of acquiring an additional unit of utility for household i. We can use the marginal price index P_i to characterize the optimal discrete choice in each sector. Below, we omit the sector subscript for ease of notation. Household i chooses the product k from firm j that maximizes

$$\{j_i, k_i\} = \underset{\{j,k\}}{\text{argmax}} \quad \ln(q_{jk}) - \frac{p_{jk}}{P_i} + \eta_j + \frac{\varepsilon_{ij}}{\sigma - 1}. \tag{2.4}$$

As we prove in the Appendix, the continuum of sectors implies that idiosyncratic tastes do not result in ex-post heterogeneity in overall consumption conditional on household income. In particular, we denote by P_l and P_h the marginal price index of low- and high-income households, respectively.

Firms. Sectors are symmetric. In each sector s, there is a fixed number N of firms. Firms differ in their brand value η_j, but are otherwise identical. The cost of producing a good is linear in the quality offered: producing a good of quality q costs κq units of labor. Each firm decides the qualities and prices of the products it sells. We restrict attention to deterministic mechanisms and assume that firms must offer their menu of products to all customers.[2] Thus, firms must design their product menu such that each consumer self-selects into their designated bundle.

There are two dimensions of consumer heterogeneity. First, consumers differ in their income level, which affects their marginal price index. Second, consumers differ in their idiosyncratic taste for brands. Idiosyncratic taste shocks affect which firm households choose to buy from, but do not affect

[1] The parameter σ, as we show below, governs the degree of substitutability across firms.

[2] The restriction to deterministic mechanisms implies firms are not allowed to sell lottery tickets to obtain their products.

their valuation of quality conditional on purchasing from a given firm. As a result, firms optimally offer only two products $k \in \{l, h\}$: one for high-income, and one for low-income consumers.

Let q_l be the product the firm offers to low-income consumers and p_l its price. $\{q_h, p_h\}$ denotes the corresponding product and price for high-income consumers. The profit maximization problem of a firm with brand value η_j is given by

$$
\begin{aligned}
\pi_j = \quad & \max_{\{q_l, q_h, p_l, p_h, B_{jl}, B_{jh}\}} \quad \frac{1}{2}\Big[B_{jl}\left(p_l - \kappa q_l\right) + B_{jh}\left(p_h - \kappa q_h\right)\Big], \\
& \text{s.t.} \qquad B_{jl} = \mathcal{B}\left(q_l, p_l, \eta_j; \{q_{j'l}, p_{j'l}, \eta_{j'}\}_{j' \neq j}, P_l\right), \\
& \qquad\qquad B_{jh} = \mathcal{B}\left(q_h, p_h, \eta_j; \{q_{j'h}, p_{j'h}, \eta_{j'}\}_{j' \neq j}, P_h\right), \\
& \qquad\qquad \ln(q_h) - \ln(q_l) \geq \frac{p_h - p_l}{P_h}, \\
& \qquad\qquad \ln(q_h) - \ln(q_l) \leq \frac{p_h - p_l}{P_l},
\end{aligned}
\tag{2.5}
$$

where the first two constraints are the customer bases as a function of product offering and market conditions, and the last two are the incentive compatibility of high- and low-income households, respectively. The customer base functions are defined as market shares within each income group, rather than as the total mass of households. That is, a firm with market share B_{jl} serves a mass of $\frac{1}{2}B_{jl}$ low-income consumers.

The incentive compatibility constraints ensure each customer is better off choosing their bundle. The high-income consumer must prefer purchasing the high-quality product to purchasing the low-quality one and using the extra funds in other sectors. The value of additional funds is given by the inverse of the marginal price index P_h. Similarly, the low-income household does not find it beneficial to switch to the high-quality product and pay the price difference, which they value at P_l.

Leveraging the Gumbel distributional assumption, the following Lemma derives an analytic characterization for the customer base functions.

Lemma **1.** *The customer base function is given by*

$$
\mathcal{B}\left(q_j, p_j, \eta_j; \{q_{j'}, p_{j'}, \eta_{j'}\}_{j' \neq j}, P\right) = \frac{e^{(\sigma-1)\left(\ln(q_j) - \frac{p_j}{P} + \eta_j\right)}}{\sum_{j'} e^{(\sigma-1)\left(\ln(q_{j'}) - \frac{p_{j'}}{P} + \eta_{j'}\right)}}. \tag{2.6}
$$

2.2 Equilibrium

We restrict attention to a symmetric equilibrium.[3] An equilibrium is a set of firm-level prices and product qualities $\{p_{jl}, p_{jh}, q_{jl}, q_{jh}\}_j$, market share distributions $\{B_{jl}, B_{jh}\}_j$, firm-level and aggregate profits $\{\{\pi_j\}_j, \Pi\}$, income levels $\{E_l, E_h\}$, and marginal price indices $\{P_l, P_h\}$, such that

1. Given their income E_i, realization of taste shocks ε_{ij}, and prices and qualities offered, each household chooses the optimal firm and quality level in each sector, as defined by (2.4), and the

[3] In Section 5, we quantify the model to match the US economy and relax the assumption of symmetry.

household problem (2.3) is solved.

2. Given the marginal price indices P_h and P_l as well as other firms' product offerings and prices $\{p_{j'l}, p_{j'h}, q_{j'l}, q_{j'h}\}_{j' \neq j}$, product offerings, market shares, and prices solve the firm's problem (2.5) for all firms.

3. The market shares $\{B_{jl}, B_{jh}\}_j$ are consistent with the market share function (2.6).

4. The marginal price indices are defined as the inverse Lagrange multipliers on the budget constraints in the household problem (2.3).

5. Aggregate profits in the economy are given by

$$\Pi = \sum_j \pi_j. \tag{2.7}$$

6. The income of each household is equal to their labor earnings plus their share of aggregate profits:

$$E_l = L_l + (1 - \alpha)\Pi, \tag{2.8}$$

$$E_h = L_h + (1 + \alpha)\Pi. \tag{2.9}$$

7. The labor market clears:

$$\kappa \sum_j \frac{1}{2} \left[B_{jl} q_{jl} + B_{jh} q_{jh}\right] = 1. \tag{2.10}$$

2.3 Welfare, Discrete Choice, and the Entropy Measure of Concentration

We now discuss properties of the equilibrium, including the determinants of welfare. We start by characterizing household optimal behavior. Because there is a continuum of symmetric sectors, the idiosyncratic taste shocks wash out in the aggregate, and all households of the same income level have the same level of aggregate consumption. Let $\tau \in \{l, h\}$ denote the income-type of households. The following proposition shows that the household problem can be solved in two stages.

Proposition 1. *The household problem (2.3) is equivalent to a two-stage problem in which the household first chooses the fraction of sectors $B_{j\tau}$ in which to purchase from each firm j, and then assigns sectors to firms.*

The average idiosyncratic taste of the chosen products in the optimal assignment equals $\sum_j B_{j\tau} \ln(1/B_{j\tau}) + \gamma$, where γ is the Euler-Mascheroni constant. The problem of a household of type τ reduces to

$$\begin{aligned} \ln C_\tau = \max_{\{B_{j\tau} \geq 0\}} \quad & \sum_j B_{j\tau} \left(\ln q_{j\tau} + \eta_j\right) + \frac{\sum_j B_{j\tau} \ln(1/B_{j\tau})}{\sigma - 1} + \frac{\gamma}{\sigma - 1} \\ s.t. \quad & \sum_j B_{j\tau}\, p_{j\tau} = E_\tau, \\ & \sum_j B_{j\tau} = 1. \end{aligned} \tag{2.11}$$

All proofs are relegated to Appendix A. The key result of Proposition 1 is showing how the chosen market shares for each firm, $B_{j\tau}$, translate, under optimal assignment, into the average taste toward the products purchased from that firm. The objective of the household problem (2.11) contains three terms. The first term is a weighted average of the quality and brand value of the products a household consumes, weighted by $B_{j\tau}$. The second and third terms capture the utility from the idiosyncratic taste shocks. The third term equals the utility a household would derive from the idiosyncratic taste shocks if there were only a single firm—γ is the mean of the standard Gumbel distribution.

The marginal price index, P_τ, can be recovered as the inverse of the Lagrange multiplier on the budget constraint in the household problem (2.11). Finally, by plugging in the optimally chosen market shares, $B_{j\tau}$, Proposition 1 yields an implicit characterization of the utility of a household of type τ,

$$\ln(C_\tau) = \sum_j B_{j\tau} \left(\ln q_{j\tau} + \eta_j\right) + \frac{\sum_j B_{j\tau} \ln(1/B_{j\tau})}{\sigma - 1} + \frac{\gamma}{\sigma - 1}. \tag{2.12}$$

Our preferences feature love-of-variety: more choices allow the household to choose a product towards which they have a higher taste. How much the availability of an additional brand increases consumer welfare depends on how often it is actually chosen, which is summarized by its market share. Equation (2.12) shows that a sufficient statistic capturing the utility derived from idiosyncratic taste shocks is the entropy measure of concentration, $\sum_j B_{j\tau} \ln(1/B_{j\tau})$. This concentration measure was introduced in Finkelstein and Friedberg (1967), and to our knowledge, ours is the first paper that derives a micro-foundation for its direct effect on welfare. The term "entropy measure of concentration" may be misleading, as higher concentration reduces this measure. Rather, it is more appropriate to call it the "entropy measure of competition." When there is only one firm to choose from, the entropy measure equals zero. A larger number of firms or lower dispersion in brand value increases the entropy measure and increases the expected idiosyncratic taste shock from the chosen brand.

2.4 Perfect Competition

So far, we have assumed that each brand j is owned by a single firm, giving rise to market power. In this section, we relax this assumption and study the perfectly competitive equilibrium. We consider the case in which each brand can be operated by a continuum of potential entrepreneurs with free entry.[4] Competition among entrepreneurs operating the same brand drives prices to marginal cost. The following proposition characterizes the resulting allocation.

Proposition **2.** *In the perfectly competitive allocation:*

(i) Quality levels are identical across brands and are given by

$$q_\tau^{PC} = \frac{L_\tau}{\kappa}, \qquad \forall \tau \in \{l, h\}, \tag{2.13}$$

[4]We assume that both the common brand value, η_j, and the idiosyncratic brand tastes, ϵ_{ij}, do not vary across potential entrepreneurs operating the same brand. Thus, entrepreneurs operating a given brand are perfect substitutes from the household's perspective: there is no love of variety within a brand, and hence no gain from having more than one entrepreneur sell the same brand in equilibrium.

(ii) Market shares of each brand j are identical across the two types of consumer segments and are given by

$$B_{j\tau}^{PC} = \frac{e^{(\sigma-1)\eta_j}}{\sum_{j'} e^{(\sigma-1)\eta_{j'}}}, \qquad \forall \tau \in \{l, h\}. \tag{2.14}$$

Proposition 2 shows that only two quality levels are produced in equilibrium: one for high-income households and one for low-income households. This result follows from two features of the model. First, all brands have the same cost of producing quality, κ. Second, the only dimension of household heterogeneity that affects the marginal valuation of quality is income. Both common and idiosyncratic brand tastes enter additively with quality. Hence, differences in quality across households are driven solely by differences in labor endowments. In equilibrium, the total amount of labor used to produce each household's consumption bundle equals that household's labor endowment.

Since quality levels are the same for all firms, households sort to brands according to their brand value, taking into account both the common component η_j as well as each household's idiosyncratic tastes ε_{ij}. Each brand's market share is a function of its brand value η_j relative to its competitors'—idiosyncratic taste shocks are i.i.d. and wash out across the continuum of consumers.

In the perfectly competitive allocation, the market share of brand j is identical in the low- and high-income segments of the population. The only difference across income types is in the quality levels consumed, which depend on each household's labor endowment. As a result, consumption inequality is equal to income inequality, as the following Lemma states formally.

LEMMA **2.** *In the perfectly competitive allocation, consumption inequality is equal to income inequality. That is,*

$$\ln C_h^{PC} - \ln C_l^{PC} = \ln L_h - \ln L_l = \ln \left(\frac{1+\alpha}{1-\alpha} \right). \tag{2.15}$$

Finally, it is worth noting that the perfectly competitive allocation is Pareto efficient. There is no misallocation between households and brands and there is no misallocation of production inputs across firms. This is, of course, not surprising given the perfect competition. But it implies that the perfectly competitive allocation is a natural baseline to compare against our benchmark equilibrium with market power.

3 Distortions in product design, misallocation, and market power

In this section, we analyze the market equilibrium and discuss inefficiencies in the model. We define as misallocation any difference between the market allocation and the perfectly competitive (PC) benchmark discussed in the previous section.

We do so in two steps. We start by considering an *island economy* in which high- and low-income consumers are segregated, so consumer heterogeneity has no direct impact on firms' choices. In this economy, we show that there is misallocation across firms due to imperfect competition: the market share of high-quality firms is too low. However, product design remains efficient, and there are no differential distortions across income groups: although consumption is lower for both groups, inequality is identical to that of the perfectly competitive allocation.

We then characterize the full market equilibrium. In the absence of sufficient competition, firms *distort* their product design choices to ensure that consumers self-select into their respective allocations. Low-income consumers are sold products of too low quality, while high-income consumers are sold products of too high quality. Further, low-income consumers are disproportionately excluded from high-quality firms. Taken together, consumption inequality is higher than in the perfectly competitive allocation.

3.1 Island economy

We start by analyzing the island economy. The firm's problem is identical to (2.5) but without the last two constraints that ensure incentive compatibility. Hence, firms effectively solve two separate profit-maximization problems, one for each consumer type. The problem of firm j in the island of consumers of type τ is given by

$$\begin{aligned} &\max_{\{p_{j\tau}, q_{j\tau}, B_{j\tau}\}} B_{j\tau}(p_{j\tau} - \kappa q_{j\tau}), \\ &\text{s.t.} \qquad B_{j\tau} = \frac{e^{(\sigma-1)\left(\ln(q_{j\tau}) - \frac{p_{j\tau}}{P_\tau} + \eta_j\right)}}{\sum_{j'} e^{(\sigma-1)\left(\ln(q_{j\tau'}) - \frac{p_{j\tau'}}{P_\tau} + \eta_{j'}\right)}}. \end{aligned} \tag{3.1}$$

Optimal product quality $q_{j\tau}$ and price $p_{j\tau}$ are given by

$$q_{j\tau} = \frac{P_\tau}{\kappa} \tag{3.2}$$

$$p_{j\tau} = \kappa q_{j\tau} \left(\frac{\sigma}{\sigma - 1} + \frac{1}{\sigma - 1} \frac{B_{j\tau}}{1 - B_{j\tau}} \right). \tag{3.3}$$

Firms choose the quality level to maximize consumer surplus, equating the marginal utility ($\frac{1}{q_{j\tau}}$) to the real marginal cost of producing quality ($\frac{\kappa}{P_\tau}$). Since taste shocks are additive, this level of quality is the same for households of the same income level.

When setting its price, the firm trades off attracting additional customers through a low price against per-customer profits. The optimal markup set by the firm, $\frac{p_{j\tau}}{\kappa q_{j\tau}}$, consists of two components. The first component is the classic monopolistic markup $\frac{\sigma}{\sigma-1}$ that arises when firms are atomistic. The second component depends positively on the market share of the firm in that consumer segment, $B_{j\tau}$. As a firm's market share grows, the potential customers it can attract by lowering its price, $1 - B_{j\tau}$, shrink. As a result, firms with larger market shares face smaller demand elasticities, and they charge a higher markup.

To close the economy, we maintain the assumption of a single labor market (Equation 2.10) and that aggregate profits (across the two islands) are split between high- and low-income consumers in proportion to their labor endowment. The following proposition characterizes the resulting equilibrium allocation in the island economy.

PROPOSITION **3.** *There is a unique equilibrium in the island economy whose allocation features*

(i) quality levels that are identical across firms, given by

$$q_\tau^{IE} = \frac{L_\tau}{\kappa}, \qquad \forall \tau \in \{l, h\}, \tag{3.4}$$

(ii) market shares of each firm j that are identical across the two types of consumer segments, implicitly defined by

$$B_{j\tau}^{IE} = \frac{e^{(\sigma-1)\eta_j - \frac{1}{1-B_{j\tau}^{IE}}}}{\sum_{j'} e^{(\sigma-1)\eta_{j'} - \frac{1}{1-B_{j'\tau}^{IE}}}}, \qquad \forall \tau \in \{l, h\}. \tag{3.5}$$

Proposition 3 shows that quality levels in the island economy are identical to the perfectly competitive allocation characterized in Proposition 2. The firms' optimality condition (3.2) implies that all firms sell the same level of quality on an island. Proposition 3 further establishes that there is no cross-subsidization in the island economy, so that the quality levels are not only constant within each island but also identical to the perfectly competitive allocation.

The second part of Proposition 3 shows that market shares of firms are, in general, different from the perfectly competitive allocation. Since high-brand-value firms charge more for the same quality level, their market share is inefficiently low. The following corollary formalizes this result.

LEMMA **3.** *There is misallocation of consumers across firms in the island economy, with high-brand-value firms serving too few, and low-brand-value firms serving too many consumers. Formally, for any two firms j and k, with $\eta_j > \eta_k$, we have*

$$\frac{B_{j\tau}^{IE}}{B_{j\tau}^{PC}} < \frac{B_{k\tau}^{IE}}{B_{k\tau}^{PC}}. \tag{3.6}$$

This result is akin to the distortion resulting from oligopolistic competition in Atkeson and Burstein (2008). Larger firms face lower demand elasticities, charge higher markups, and end up selling less than they would under perfect competition. While misallocation reduces aggregate consumption for each type of household, the following Proposition shows that this proportional reduction in consumption is identical across the two consumer types. That is, the distortion in market shares is *independent* of consumer income: In both the high- and low-income islands, market shares and hence misallocation across consumers are identical.

PROPOSITION **4.** *Consumption is inefficiently low in the island economy, but consumption inequality* *is identical to consumption inequality in the perfectly competitive allocation. That is,*

$$\ln C_\tau^{IE} \leq \ln C_\tau^{PC}, \qquad \forall \tau \in \{l, h\} \tag{3.7}$$

$$\ln C_h^{IE} - \ln C_l^{IE} = \ln C_h^{PC} - \ln C_l^{PC}. \tag{3.8}$$

Towards a market equilibrium. In the island economy, we relaxed the two incentive compatibility constraints:

$$\ln(q_{jh}) - \ln(q_{jl}) \geq \frac{p_{jh} - p_{jl}}{P_h} \tag{3.9}$$

$$\ln(q_{jh}) - \ln(q_{jl}) \leq \frac{p_{jh} - p_{jl}}{P_l} \tag{3.10}$$

These ensure that each consumer prefers choosing their designated product over that of the other income group. Before turning to the analysis of the market equilibrium, it is useful to examine which consumers have an incentive to purchase from the other island, and which brands they are incentivized to purchase. Note that if no incentive compatibility constraint is violated, the market equilibrium allocation is identical to the island economy one.

The following proposition characterizes which incentive compatibility constraints are violated in the island economy allocation.

Lemma **4.** *In the island economy allocation, the incentive compatibility (IC) constraints of low-income households are slack. The IC constraints of high-income households are violated for the top $\widetilde{N}$ brands. That is, if the high-income IC constraint is violated for a brand with value $\bar{\eta}$, it is violated for all brands with $\eta \geq \bar{\eta}$.*

In the island economy, firms charge positive markups. At the unit price they face, all consumers would prefer to purchase only a fraction of the quality they are offered. However, since goods are indivisible, the only possible deviation available to consumers is to buy the product designed for the other income group. Low-income consumers never want to deviate, because they would like to reduce quality and the alternative product instead offers higher quality.

High-income consumers trade off a lower cost associated with the low-quality bundle with the utility loss due to the lower quality of the good. Because quality is equated across firms within each consumer type, the utility loss associated with downgrading a product is the same across all brands. However, heterogeneous markups imply that the cost savings from downgrading a high-brand-value firm is larger. As a result, there is a cutoff brand value above which high-income households would like to downgrade their products and below which the incentive compatibility is slack.

3.2 Market allocation

We next analyze the market allocation, where firms must design their product offerings so that each consumer purchases their designated bundle. We start by characterizing firms' optimal allocations, taking price indices as given. We then turn to study the role of market power in shaping consumption inequality in general equilibrium in a special case of the general model.

As Lemma 4 shows, the allocations in the island economy may violate the incentive compatibility (IC) constraints of high-income consumers for brands with sufficiently high market power. Proposition 5 characterizes the firm's optimal choices in the region in which the IC constraints are slack and the region in which the high-income IC constraint is binding.[5]

Proposition **5.** *The firm's optimal policy in the unconstrained region as well as the region where the IC of the high type binds is given by the following expressions.*

(i) Optimal quality levels are

$$q_{jh} = \frac{P_h}{\kappa}, \tag{3.11}$$

$$q_{jl} = \frac{P_l}{\kappa} - \psi \left(\frac{1 - \frac{P_l}{P_h}}{\kappa B_{jl}} \right), \tag{3.12}$$

(ii) Optimal prices are

$$p_{jh} = \kappa q_{jh} \tilde{\mu}_{jh} - \psi \frac{1}{B_{jh}} \left(\tilde{\mu}_{jh} - 1 \right), \tag{3.13}$$

$$p_{jl} = \kappa q_{jl} \tilde{\mu}_{jl} + \psi \frac{1}{B_{jl}} \left(\tilde{\mu}_{jl} - 1 \right), \tag{3.14}$$

where $\psi \geq 0$ is the Lagrange multiplier on the high-income IC constraint and $\tilde{\mu}_{j\tau} \equiv \left(\frac{\sigma}{\sigma-1} + \frac{1}{\sigma-1} \frac{B_{j\tau}}{1-B_{j\tau}} \right)$ is the unconstrained markup.

If the IC constraint is slack ($\psi = 0$), firm j offers the same bundle as it would in the island economy allocation *conditional* on the two marginal price indices. Both high- and low-income consumers are offered a quality level that equates their marginal utility to the real marginal cost. Note that, as long as the IC binds for some firm in the economy, P_h and P_l, and hence equilibrium quality levels offered are different from the island economy allocation. When $\psi = 0$, the price a firm charges also simplifies to the familiar $p_{j\tau} = \kappa q_{j\tau} \left(\frac{\sigma}{\sigma-1} + \frac{1}{\sigma-1} \frac{B_{j\tau}}{1-B_{j\tau}} \right)$. Markups may still differ across the two consumer segments, since firms do not necessarily have equal market shares in both.

Firms for which the IC is binding cannot design their products for each income group in isolation. Instead, when designing their product portfolios, firms must take potential cannibalization into account—how increasing the surplus for one income group affects the surplus they must provide the other income group. If the bundle offered to low-income consumers is too attractive relative to the more expensive high-quality bundle, high-income consumers will not purchase their designated product. To ensure incentive compatibility, the firm has four margins it can adjust: the two quality levels and the two prices charged.

On the quality margin, firms choose to distort only the low-quality product but not the high-quality one. The quality level of each good in the island economy was chosen to maximize the joint

[5] In theory, there could be an additional region in which the incentive compatibility constraint of low-income households is binding. However, we confirm numerically that in general equilibrium there is no parameter configuration in which that is the case for any firm.

surplus, $\ln(q_{j\tau}) + \eta_j - \frac{\kappa q j \tau}{P \tau}$. There is no reason for the firm to reduce the joint surplus for high-income consumers, as that would not alleviate the incentive compatibility constraint. On the other hand, reducing the surplus delivered to low-income consumers helps alleviate the IC constraint. As a result, the firm chooses to distort the quality level of the low-end product downwards.

The degree of quality distortion is governed by three variables, as shown in equation (3.12). First, the larger the Lagrange multiplier, ψ, the larger is the distortion. Second, the distortion is decreasing in the ratio of aggregate price indices, $\frac{P_l}{P_h}$. This ratio governs the relative valuation of quality of high- vs. low-income consumers. The larger this gap in valuations, the larger the difference in willingness to pay for quality between the two consumer segments. As long as this gap is positive, reducing quality of the low-end product hurts rich consumers more than poor ones. Third, the distortion is decreasing in the market share of low-income consumers. The higher the sales to low-income consumers, the higher the cost of distorting their quality.

While the incentive compatibility constraints push down the quality of low-quality goods produced by high-market-power firms, a general equilibrium force operates in the opposite direction. If all quality levels either stay the same or go down, aggregate labor demand would not meet supply, and the labor market would not clear. In general equilibrium, labor market clearing requires a decline in the real wages—the inverse of the aggregate price indices. Lower real wages, in turn, raise equilibrium quality choices uniformly across firms. As a result, the equilibrium quality of high-quality goods exceeds its first-best level. For low-quality goods, the net effect is heterogeneous: quality rises for firms with sufficiently low market power, but declines for firms with high market power.

On the price margin, firms choose to operate on both fronts: reducing the markup charged to high-income consumers and raising it for low-income consumers. Optimal prices now balance two motives. The first part of Equations (3.13) and (3.14) is the same as in the island economy: firms choose prices that trade off attracting additional customers and sales per customer. The second part of the optimal prices reflects the need to preserve incentive compatibility.

The magnitude of price distortion depends not only on the Lagrange multiplier, ψ, but also on the market share and demand elasticity in each consumer segment. The larger the market share and the demand elasticity, the larger the profit losses from price distortions. As a result, both these forces incentivize the firms to use other margins to satisfy the incentive compatibility constraint.

Relative to the island economy, welfare of each consumer type changes along two margins. First, as discussed above, high-income households get allocated a higher quality of each good they consume. Low-income households, on the other hand, are allocated, on average, less of each good. Second, firms for which the IC of the high type is binding distort prices and allocations, which results in a change in market shares. Lower prices for the rich attract more customers to these firms, while higher prices and lower quality levels for the poor lead to lower market shares in that consumer segment. Because the market share of high-market-power firms in the island economy is distorted downwards (Lemma 3), the reallocation of market shares toward high-market-power firms benefits the rich, while the reallocation of market shares away from high-market-power firms hurts the poor.

3.3 Market power and consumption inequality

There are two sources of market power in the economy: the number of competitors in a sector and the dispersion in brand value. To understand how market power shapes consumption inequality, we consider a special case of the general model and analyze the comparative statics with respect to each of these sources of market power in isolation.

We consider an economy in which $N-1$ firms have the same brand value, which we normalize to zero, while a single firm has brand value $\eta = \Delta > 0$. We refer to these firms as the *small* firms and the *large* firm, respectively. This specification captures both sources of market power through two parameters: the number of firms N and the brand-value advantage of the large firm Δ. As in the analysis above, the comparative statics consider changes to N and Δ that are identical across all sectors.

We begin with the comparative statics with respect to the number of firms. To isolate the role of the number of competitors, we consider the limit in which the advantage of the large firm vanishes, $\Delta \to 0$. In this limit, firms are effectively symmetric: each firm holds a market share of $1/N$ in both consumer segments, and the degree of competition is governed solely by the number of firms. The following proposition establishes the link between the number of competitors and consumption inequality.

PROPOSITION **6.** *In the limit as $\Delta \to 0$, consumption inequality, $\ln C_h - \ln C_l$, is weakly decreasing in the number of firms N.*

With fewer firms, the market share of each firm rises, and with it the markup it optimally charges. Higher markups widen the price gap between high- and low-end products, which strengthens the incentive of high-income consumers to purchase low-end products. When the number of firms is small enough, the IC constraint binds for every firm in the sector. Facing cannibalization concerns, firms distort the quality of their low-end products downward, raise prices at the bottom, and reduce them at the top. Because firms are symmetric, low-income consumers cannot substitute toward less-distorted competitors, and their consumption falls across the board. In general equilibrium, labor market clearing requires that the consumption of high-income households rise: some of the labor endowment of low-income households is used to produce goods consumed by high-income ones. As a result, consumption inequality increases as the number of firms falls.

We next turn to the second source of market power: the dispersion in brand value across firms. To characterize its effect analytically, we focus on the case of two firms, $N = 2$, and ask how the brand-value advantage of the large firm, Δ, shapes consumption inequality. A rise in Δ raises the market share of the large firm and thereby concentrates market power in its hands. The following proposition shows that consumption inequality is minimized when brand-value dispersion vanishes: any brand-value dispersion can only raise consumption inequality relative to the symmetric economy.

PROPOSITION **7.** *Suppose $N = 2$. Consumption inequality, $\ln C_h - \ln C_l$, is minimized in the limit as the brand-value advantage of the large firm vanishes. That is, for any $\Delta > 0$, consumption inequality is weakly higher than in the limit $\Delta \to 0$.*

A brand-value advantage raises the large firm's market share for any given quality-price bundle. With a larger market share, the large firm faces a lower demand elasticity and optimally charges higher markups. Higher markups, in turn, increase the scope for within-firm cannibalization, as high-income consumers are tempted to switch to the firm's low-end product. Consequently, the large firm distorts the quality of its low-end product downward and raises its price, while reducing the price of its high-end product. Faced with lower quality at a higher price, low-income consumers increasingly purchase from the small firm, worsening the misallocation of consumers across firms in their segment. High-income consumers, in contrast, are drawn to the large firm by its lower high-end price, which undoes some of the baseline misallocation in their segment. Both margins—the widening quality gap and the asymmetric reallocation of consumers across firms—raise consumption inequality relative to the symmetric economy.

3.4 Discussion

Before turning to the empirical evidence, we discuss some of the assumptions of our model and their relevance for the main results.

First, firms differ only in their brand value but have identical marginal production costs. Alternatively, one could assume that firms differ in their cost of production, κ_j, but have identical brand values η. Given our assumptions of constant returns to scale and log-additive preferences in quality and brand value, these two formulations are isomorphic. Defining a firm's effective brand value as $\ln(1/\kappa_j)$, one can rewrite the entire problem analogously to the baseline formulations, and all propositions continue to hold.

Second, the economy consists of only two income types, high and low. This assumption is primarily imposed for tractability and expositional purposes. In an environment with a continuum of income types, in which firms offer a continuous menu of products, our perfect competition and island economy results carry over: pairwise consumption inequality mirrors income inequality. The key difference lies in the set of incentive compatibility constraints. Instead of a single occasionally binding constraint per firm, there will be a continuum of these constraints. With standard regularity conditions on the income distribution, these ICs would all be slack but for the locally downward ones. That is, each household would be indifferent between their designated product and the product designed for the household with infinitesimally less income. While the continuum of these locally downward ICs makes the problem more challenging to solve, we do not believe it will change the main mechanism we study: the downward quality distortion for low-income types will be increasing in the degree of market power of the brand.

Third, our model has the stark prediction that there is perfect separation by types: all high-income consumers purchase the high-income product and vice-versa. This result is driven by the assumption that taste shocks vary only across brands and not across products within a brand. In principle, one could introduce an additional taste shock over the quality of a specific good. Instead of a standard IC, there would be a mapping from prices and qualities offered to the fraction of high and low-income consumers buying each product. The economics are likely to remain similar, but extending the model in this direction requires a set of non-trivial modeling assumptions. For example, if these taste shocks are i.i.d. over each quality type, firms would have an incentive to offer an infinite number of qualities on a small interval.

4 Empirical Evidence

The theory developed in the previous section delivers two main empirical predictions. First, firms with sufficient market power distort the quality of their low-end products downward, not because producing higher quality is prohibitively costly, but because doing so would cannibalize sales of their high-end offerings. Second, more concentrated markets exhibit greater dispersion in firms' market shares across income groups: dominant firms capture a disproportionate share of high-income consumers while losing low-income consumers to smaller competitors.

We present evidence for each of these predictions in turn. We first discuss examples, drawn from historical and contemporary markets, in which firms with market power have deliberately degraded the quality of low-end products to segment consumers by willingness to pay. We then turn to a systematic analysis using the NielsenIQ Consumer Panel dataset, which allows us to test whether the predicted relationship between market concentration and the income composition of firms' customer bases holds across a broad set of product categories.

4.1 Historical and contemporary examples

The mechanism at the center of our model, that firms with market power deliberately degrade the quality of their low-end products, has a long history in practice. Before turning to systematic evidence, we briefly discuss examples that illustrate this mechanism. The main goal of this subsection is not to establish that quality degradation occurs in practice, as the examples below have been previously discussed in the literature. Rather, we highlight that these classic examples all arise in markets with limited competition.

Railroad Transportation in 19th Century Britain. In the early decades of the British railway system, each line required its own Act of Parliament and was built and operated by a single chartered company. By the early 1840s there were over one hundred such companies, each holding a monopoly on its own route (McLean and Foster, 1992). Passengers on these lines could choose among three classes of carriage. First-class carriages were fully enclosed, padded, and relatively comfortable. Second-class carriages were roofed but sparsely upholstered. Third-class passengers, by contrast, rode in open wagons with no roof, no glazing, and wooden benches.

The conditions were not a reflection of cost considerations by railway companies given third-class passengers' willingness to pay. Dupuit (1849), a French civil engineer who observed the same practice in the French railway system, argued that the reason was instead market segmentation:[6]

[6]Our translation. Original French (p. 234): "*Ce n'est pas à cause de quelques milliers de francs qu'il en coûterait pour mettre un toit sur les wagons de troisième classe ou pour rembourrer les banquettes de troisième classe, que telle compagnie a des wagons découverts avec des bancs de bois [...] l'on frappe sur le pauvre, non pas qu'on ait envie de le faire souffrir personnellement, mais pour faire peur au riche.*"

> "It is not because of the few thousand francs it would cost to put a roof over the third-class wagons or to upholster their seats that some company or other has open wagons with wooden benches [...] one strikes at the poor, not out of a desire to make them suffer, but to frighten the rich."

Through the lens of our model, the railway companies faced no competitive pressure on their respective routes and therefore distorted the quality of their low-end product to prevent cannibalization of first- and second-class revenue.

The British government ultimately intervened. The Railway Regulation Act of 1844, mandated that every company run at least one train per day in which third-class passengers were "protected from the weather and provided with seats," at an average speed of at least 12 miles per hour and a fare of at most one penny per mile (McLean and Foster, 1992). Railway companies complied with the letter of the law but not its spirit. They scheduled the mandated "parliamentary trains" at the least convenient hours, used the most uncomfortable carriages the law would allow, and forced passengers to wait hours at intermediate stations. *Lloyd's Weekly Newspaper* described these trains in 1852 as "as uncomfortable as the law will allow," running at "a pace that wearies out all patience" (Wolmar, 2007). Lord Carlingford later admitted in Parliament that the result was "virtually a fourth class of passenger carriage" that was "in all other respects objectionable." Even when regulation tried to impose a minimum quality floor, monopolists found margins along which to degrade service. The lesson extends beyond railways: when quality distortions are rooted in market structure, regulation may be a poor substitute for competition.

Hardware Crippleware. The same mechanism operates in contemporary technology markets, where it has acquired a name: *hardware crippleware*. The term refers to products in which a manufacturer deliberately disables features of a physically capable device to create a lower-tier version.[7]

A well-known example is Intel's 486SX microprocessor, introduced in April 1991. Intel entered 1991 with a near-monopoly in the 32-bit x86 microprocessor market.[8] The 486SX was physically identical to the higher-end 486DX, but with the math coprocessor disabled. Deneckere and McAfee (1996) discuss the 486SX as a canonical example of 'damaged goods,' arguing that disabling the FPU is consistent with a strategy of market segmentation: by offering a chip with reduced functionality, Intel could charge a premium for the full-featured version without losing price-sensitive customers entirely. Intel's dominance meant that cannibalization of the 486DX was a first-order concern. Disabling the coprocessor was the firm's response.

This practice is common in markets where a small number of firms hold dominant positions. In 2001, IBM introduced "Capacity on Demand" for its zSeries enterprise servers, shipping systems with processor cores physically present but inactive at delivery. Customers could later activate this dormant capacity via software keys upon payment, without replacing or modifying the hardware. In

[7] We note that hardware crippleware is not fully captured in our model, as we assume a linear quality production cost. One way to extend our model to capture such behavior is to have a step function for the production cost.

[8] Dataquest describes Intel as having held a "complete monopoly" in this market until 1991 (Lowe, 1993). Intel remained dominant through the end of the decade; the European Commission reports that its worldwide x86 CPU market share was consistently around or above 70% from 1997 onward (European Commission, 2009).

2016, Tesla began selling the Model S 60, which contained the same 75 kWh battery pack as the Model S 75 but was software-locked to 60 kWh. Customers could unlock the full capacity for an additional fee (Weintraub, 2016). In 2020, NVIDIA introduced its Ampere-based GeForce RTX 3080 and 3090 graphics processors, which were built on the same GA102 silicon but differed in the number of enabled streaming multiprocessors and memory configuration (NVIDIA Corporation, 2020).

What unites these examples, and what connects them to our theoretical framework, is the role of market power. Quality degradation of this kind is not a generic feature of multi-product firms. It arises when a firm's market position is strong enough that cannibalization of its high-end products becomes a binding concern. A competitive firm facing the same production technology would have no reason to disable a coprocessor or remove a roof from a railway carriage. Doing so would simply make it lose customers to rivals offering the undegraded product.

4.2 Evidence from grocery store goods

We now test the second main prediction of our model: the more concentrated a market is, the more the high-brand-value firms distort downwards the product offered to low-income households and reduce prices at the top. As a result, their market share among high-income households increases, while their popularity among low-income households declines. To see whether this pattern holds across a broad set of markets, we turn to the NielsenIQ Consumer Panel dataset. We use this particular setting, since it allows us to observe both the income of a household and detailed spending on all brands and products.

4.2.1 Data and Descriptives

We use the NielsenIQ Consumer Panel dataset from 2007–2020 to conduct our analysis. We focus on core grocery goods, which include Health & Beauty Care, Dry Grocery, Frozen Foods, Dairy, and Non-Food Grocery. The analogue to a market in our model is a product module, e.g., "olive oil" or "deodorants". Our sample covers 869 product modules. In each of these markets, every year, we analyze the behavior of the top firm, defined as the firm with the largest market share. For robustness, we also consider the top three firms in each market. We define high- and low-income households as those with above and below median income, respectively.

Table 1 presents descriptive statistics. The average market share of the top firm is 32.4%. This firm controls, on average, 33.3% of the high-income market segment and 31.7% of the low-income market segment. Firms offer multiple products within each module. Panel B of the table shows that the median top firm offers products with 99 distinct Universal Product Codes (UPCs) across 23 different sizes.

In our model, firms offer products of different qualities. To gauge the extent to which firms offer products of heterogeneous qualities, we proceed in two steps. First, Panel C of Table 1 shows that high-income households purchase more expensive products. On average, the products they purchase are about 10% more expensive. This difference cannot be fully explained by size variation. The

Table 1: Descriptive Statistics

	Mean	Median	S.D.
Panel A: Market Structure			
Top firm market share	0.324	0.299	0.167
Top firm market share, high-income	0.333	0.302	0.179
Top firm market share, low-income	0.317	0.296	0.161
HHI	0.185	0.151	0.136
Panel B: Product Variety (Top Firm)			
Distinct UPCs per module-year	167.2	99	251.2
Distinct sizes per module-year	35.3	23	38.7
Panel C: Size and Price (Top Firm)			
Relative log price (high − low income)	0.101	0.083	0.084
Relative log size (high − low income)	0.085	0.063	0.094
Relative log price per unit (high − low income)	0.016	0.009	0.059

Notes: The sample covers 869 product modules across 11,757 module-year observations. All statistics are weighted using the total sales in the module. Top firm is the firm with the largest overall market share in each module-year.

price they pay per unit of good, e.g., price per oz, is about 1.6% higher.[9] Second, we study the variation of product attributes within firms. 96% of sales are in module-year pairs where the top firm offers products that differ by at least one product attributes.[10] In particular, 72.4% of sales are in module-years where the top firm offers both organic and non-organic versions of their product.

4.2.2 Market concentration and relative market shares

Our key variable of interest is the relative market share of the top firm, defined as the top firm's market share among high-income households divided by its market share among low-income households. In the absence of cannibalization concerns within the firm, the relative market share would be one in our model; firms are equally popular across all income segments. The lower the external competition a firm faces, the more it tilts qualities and prices in favor of high-income consumers, increasing its market share in that segment while losing low-income customers. We therefore test whether the relative market share of the top firm varies systematically with market concentration. Our baseline measure of market concentration within each module is the Herfindahl–Hirschman Index (HHI). Our baseline regression takes the following form:

$$\log\left(\frac{s^H_{mt}}{s^L_{mt}}\right) = \delta_t + \beta \log HHI_{mt} + \epsilon_{mt}, \tag{4.1}$$

where s^H_{mt} and s^L_{mt} are the market share of the top firm in market m in year t in the high- and low-

[9] Quantity discounts are prevalent in this market, see Bornstein and Peter (2025). This feature of the data would lead to lower price-per-unit for larger sizes, indicating that high-income households indeed purchase higher quality goods.

[10] We consider the following product attributes: (i) organic claim, (ii) type, (iii) flavor, (iv) form, (v) formula, (vi) style, and (vii) scent.

income market segments, respectively. We control for time fixed effects, δ_t, and study the elasticity of the relative market share with respect to the HHI in the market.

Table 2 reports the results. Column (1) presents the benchmark specification. The elasticity of relative market shares with respect to HHI is 0.074 and is statistically significant. To interpret the economic magnitude, note that the median relative log market share is 0.042. The coefficient implies that a one standard deviation increase in log-HHI raises it from 0.042 to 0.096. Column (2) shows that the same pattern holds if we simply compare markets with above median HHI to ones below it. The difference is 8.9 log points.

Through the lens of our model, the distortions the firm introduces directly depend on its own market share. Specification (3) in Table 2 considers the same regression specification, replacing the market HHI with the firm's own market share. This specification also yields a positive and significant coefficient. The elasticity of the relative market share with respect to the firm's own market share is 0.096. That is, the relative market share of a firm with a 10% larger market share is about 1% higher.[11]

Table 2: Concentration and relative market share

	Top Firm			Top-3 Firms		
	(1)	(2)	(3)	(4)	(5)	(6)
log(HHI)	0.074***			0.037***		
	(0.021)			(0.011)		
1[HHI > median]		0.089***			0.046***	
		(0.028)			(0.014)	
log(own market share)			0.096***			0.073***
			(0.025)			(0.023)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,757	11,757	11,757	11,757	11,757	11,757
Clusters	869	869	869	869	869	869
R^2	0.089	0.069	0.087	0.116	0.098	0.117

Notes: The dependent variable is $\log(s^H_{f,m,t}/s^L_{f,m,t})$, where s^H and s^L denote the market share of firm f among high- and low-income households in module m and year t. Columns (1)–(3) use the top firm by overall market share; columns (4)–(6) use the top three firms combined. All regressions include year fixed effects and are weighted by module-year sales. Standard errors clustered by product module are in parentheses. $^{***}\ p < 0.01$; $^{**}\ p < 0.05$; $^{*}\ p < 0.10$.

Finally, columns (4–6) repeat the analysis but consider the combined market share of the top three firms instead of only the top firm. The coefficients are all positive and significant, further strengthening the hypothesis that lack of external competition tilts the market share of top firms towards the high-income segment. Moreover, consistent with the model's prediction, these coefficients are all smaller. The firms ranked 2–3 do not have as much market power as the top firm, and would therefore not be expected to distort their product portfolios as much as the top firm.

[11]Note this is not 10 percentage points higher market share, but 10 percent. i.e., a firm with a market share of 33% relative to a firm with a market share of 30%.

5 Quantitative Results

To quantify the distributional consequences of market power, we extend the model to heterogeneous sectors and calibrate it to the 2022 U.S. economy, matching the degree of competition sector by sector. We find that market power lowers low-income households' welfare by over 7.3 log points relative to the perfectly-competitive allocation and raises that of high-income households by 1.6 log points. The calibrated model allows us to decompose the welfare effects of market power into the misallocation margin and the quality margin, revealing that the majority of the welfare gap is driven by the quality margin. We then recalibrate the model to the U.S. economy in the 1990s and study the distributional impact of the rise in market power over the past three decades.

5.1 Calibration

The model environment in Section 3 features symmetric sectors. For the calibration, we consider an extension with S types of sectors. Sectors of type s each contain N_s firms; each of these firms draws its brand value from a Pareto distribution with shape χ_s. We normalize the scale of the Pareto distribution to one in each sector.[12] Households aggregate sectoral consumption through a Cobb-Douglas aggregator with weight ω_s on sectors of type s. We maintain the assumption that there are two types of households that differ by their income. Details on how we computationally solve the model are relegated to Appendix B.1.

Table 3 summarizes the calibrated parameters and targeted moments. The model has two sets of parameters: those that are common across sectors and those that are sector-specific. Two parameters are common. The first is the elasticity of substitution across brands, σ, which we set to 3 following Midrigan (2011). The second is the degree of income inequality, α, which is calibrated to 0.31 in order to match expenditure inequality in the Panel Study of Income Dynamics (PSID) in 2022. Specifically, we match the relative expenditures on non-housing consumption of households above and below median income, excluding the top and bottom 10%.[13]

The remaining parameters are sector-specific and are calibrated to the Establishment and Firm Size Statistics data tables from the 2022 Economic Census. We calibrate the model to 897 6-digit NAICS industries. We drop 2 industries for which the top 4 sales share is not available. Each industry maps to a sector type in the model. For each industry, we set N_s to the observed number of firms, the parameter ω_s to match each sector's share of total sales, and the Pareto shape of brand value dispersion χ_s to match the share of sales by the top 4 firms.[14] The model hits all targets precisely. Figure 1 presents a scatterplot of the number of firms and sales concentration across all 897 sectors of the economy.

[12] The scale of the Pareto distribution within a sector is a level shifter on brand values. It affects the level of consumption from that sector but not the within-sector distortions or relative outcomes across income groups.

[13] We match expenditure rather than income inequality, as our model is static and abstracts from savings.

[14] We target the top-4 sales share rather than sectoral HHI, since the latter is missing for 220 sectors. In Appendix B.5, we re-estimate the model targeting the HHI instead and confirm that our main conclusions hold in this version.

Table 3: Calibrated Parameters

Panel A. Common parameters				
Parameter	Description	Value	Moment	Source
σ	Dispersion of taste shocks	3	Elasticity of substitution across firms	Literature
α	Spending inequality	0.31	Non-housing exp. inequality	PSID (2022)

Panel B. Sector-specific parameters						
Parameter	Description	Mean	Median	SD	Moment	Source
N_s	Number of firms	6,952	1,189	18,544	# firms in 6-digit NAICS	Census (2022)
χ_s	Pareto shape	5.3	5.3	1.1	Top-4 sales share	Census (2022)
ω_s	Sector weight	0.001	0.0003	0.003	Sales share	Census (2022)

Notes: Mean, median, and standard deviation are computed across the 897 6-digit NAICS industries for sector-specific parameters. α is chosen to match the relative expenditures of households above and below median income, trimming the top and bottom 10% of households.

5.2 Consumption Inequality

We first study the degree of consumption inequality in the calibrated economy. Recall that absent cannibalization considerations, the degree of consumption inequality mirrors that of income inequality. Cannibalization considerations lead high-market-power firms to reduce the quality of their low-end products and raise their price. This force increases consumption inequality beyond income inequality.

Quantitatively, we find that consumption inequality is 8.9 log points larger than income inequality: 73 vs. 64 log points respectively. That is, market power widens consumption inequality by 14%. Table 4 presents our findings. Relative to the perfectly-competitive allocation, low-income households' welfare is 7.3 log points lower in the market allocation, while high-income households are 1.6 log points better off.

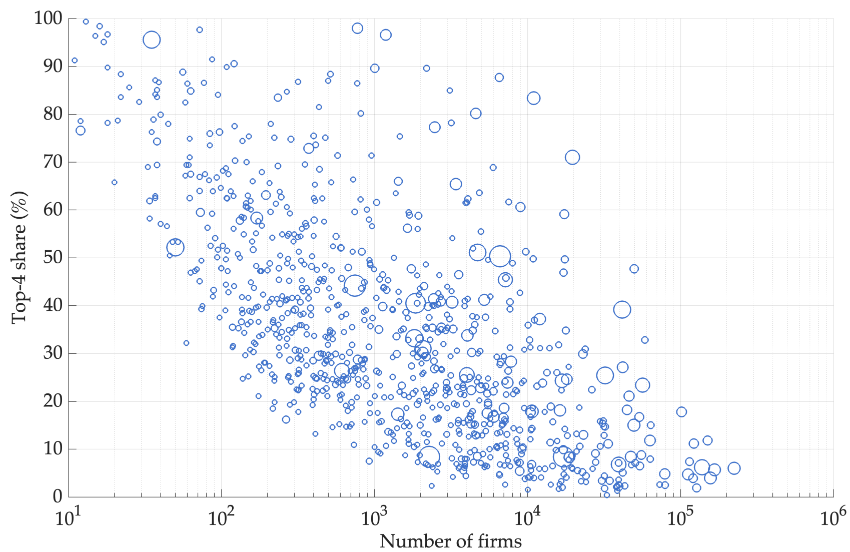
Table 4: Aggregate Consumption Relative to the Perfectly-Competitive Allocation

	High-income	Low-income	Inequality
Market equilibrium	$+1.6$	-7.3	8.9
Quality margin	$+2.4$	-5.1	7.6
Misallocation margin	-0.9	-2.2	1.4

Notes: This table reports the log points difference in aggregate consumption of each income type relative to the perfectly competitive allocation, and decomposes it into quality and misallocation margins. The first row corresponds to the market allocation. The quality margin isolates the effect of quality distortions, while the misallocation margin isolates the effect of distorted market shares.

The higher consumption inequality in the market equilibrium is driven by both quality distortions and misallocation. On the quality margin, high-brand-value firms distort the quality level of their low-end products downward. The left panel of Figure 2 illustrates this. Relative to the perfectly-competitive allocation, the quality level offered to low-income households is lower for all firms (orange line). The size of the distortion tends to increase in a firm's market share: at the 10th percentile of firm sales, the quality distortion is -2.7 log pts, increasing to -12.5 log pts at the 90th percentile and

Figure 1: Sales Concentration and Number of Firms Across Industries



Notes: This figure presents a scatterplot of the number of firms in each 6-digit NAICS industry against the sales share of the largest 4 firms. The size of each circle corresponds to the sector's overall sales share in the economy.

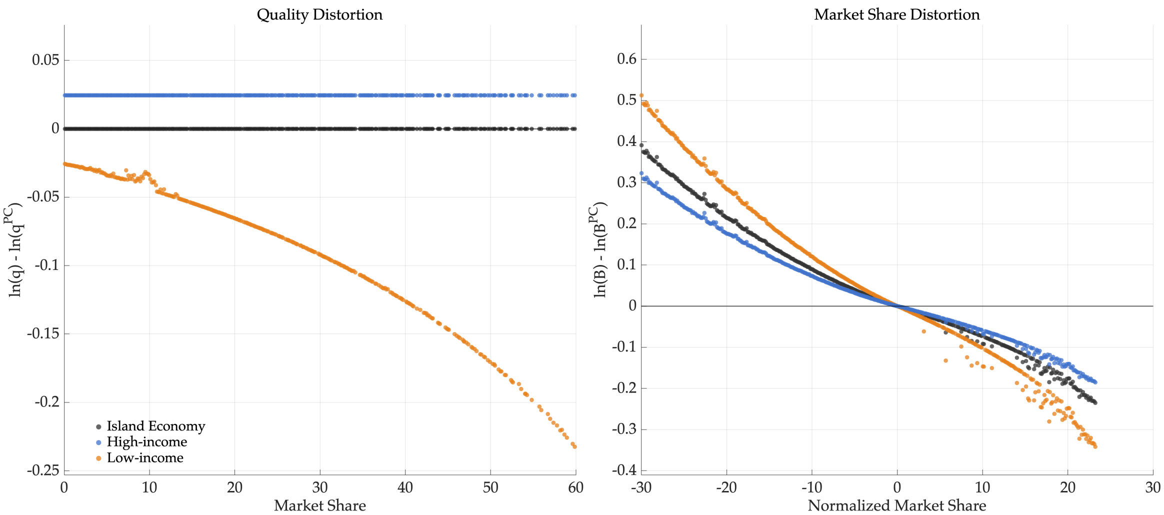
all the way to -63 log pts at the 99th percentile. On the flip side, high-income consumers are being offered 2.4 log pts higher quality across all firms (blue line).

The second channel—misallocation of consumers across firms—plays a smaller role. If there were only the cross-firm misallocation channel, consumption would be 0.9 log pts lower for the rich and 2.2 log pts lower for the poor. That is, misallocation across firms worsens for low-income households relative to the island economy, since they are increasingly driven away by high-brand-value firms, and improves for high-income ones.

The right panel of Figure 2 plots the normalized market shares of each firm relative to the perfectly-competitive allocation. A firm's market power depends on its market share *relative* to its largest competitors. For example, a firm with a 10% market share may be a small player in a very concentrated market, or a dominant player in a less concentrated one. Therefore, we define the normalized market share of a firm as its market share relative to the "undistorted" firm. The undistorted firm has exactly the level of η such that its market share coincides with the one in the perfectly-competitive allocation.

Figure 2 plots this normalized market share in the island economy as well as in the market economy for both income groups. All three lines are different from zero and downward sloping, meaning that there is misallocation of consumers across firms. High-brand-value firms, i.e., those with larger overall market shares, have too few consumers, while too many households buy from low-brand-value firms that charge lower markups. The allocation of consumers across firms improves for high-income households in the market allocation: the blue line is flatter than the island economy. For low-income households, distortions in quality and higher prices at top firms exacerbate the misallocation present in the island economy.

Figure 2: Quality distortions and market shares relative to perfectly-competitive allocation



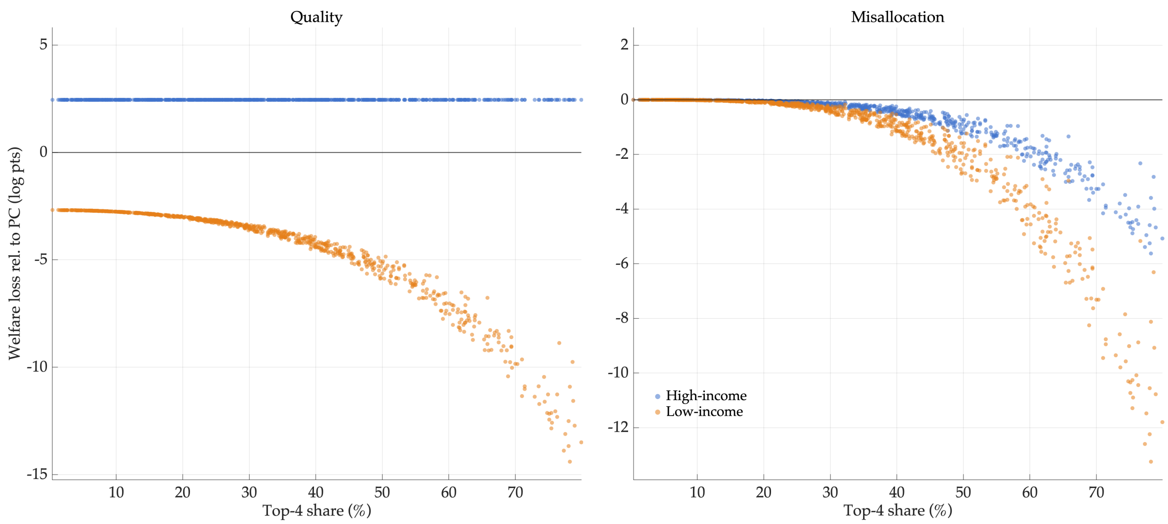
Notes: The left panel plots the log difference in quality between market and perfectly competitive allocation against a firm's market share. The x-axis is capped at 60% for visual ease. In both panels, high-income households are in blue, low-income ones in orange, and the island economy is represented by black dots. The right panel plots each firm's market share relative to the perfectly competitive allocation. Numbers are reported as log differences. The x-axis measures the firm's normalized market share, in percentage points: its market share (pooled across income segments) minus the sector-specific share at which the island-economy and high-income market series cross. The x-axis is capped at ±30 percentage points for visual ease.

Sectoral heterogeneity. In the aggregate, high-income consumers enjoy 1.6 log pts higher welfare than they would in the perfectly-competitive allocation, while low-income consumers are over 7 log pts worse off. This aggregate outcome masks considerable heterogeneity by sector. In Figure 3, we plot the two margins of consumption inequality as defined in Table 4 for each sector, as a function of sales concentration. While a sector's top 4 sales shares are not a sufficient statistic for the degree of distortion, they capture a large part of sectoral heterogeneity.

High-income consumers are always offered a 2.4 log pts higher quality level. Similar to the analysis by market share, more concentrated markets have a higher degree of downward distortion for low-income consumers. In the most competitive sectors, quality offered to the poor is about 2.7 log points lower, but this number goes up to 36 log points in the 1% most concentrated ones.

The right panel of Figure 3 plots the degree of misallocation resulting from distortions in market shares in each sector. Three takeaways emerge. First, not surprisingly, the degree of misallocation across firms is increasing in market concentration. It ranges from nearly zero in competitive markets to more than 10 log pts welfare loss in the most concentrated markets. Second, misallocation always hurts the low-income households more. Third, some markets are so concentrated that the losses from misallocation across firms for high-income consumers outweigh the gains from higher levels of quality. This illustrates that our aggregate result that high-income consumers are better off in the market allocation hinges on the level of concentration: in industries that are very far from competitive, even high-income consumers could be worse off than in the perfectly-competitive allocation.

Figure 3: Market concentration and the two channels of consumption inequality



Notes: This figure plots each sector's welfare loss due to quality distortions (left panel) and misallocation (right panel) for high-income consumers in blue, and low-income consumers in orange. Losses are expressed in log points of sectoral consumption, relative to the perfectly-competitive allocation. The x-axis measures the sector's share of sales accounted for by the largest 4 firms and is capped at 80% for visual ease.

5.3 Distributional impact of the rise of market power

Since the 1990s, there has been a secular rise in market concentration in the U.S. To gauge whether this rise differentially affected households of different income groups, we recalibrate the model to match the level of concentration 30 years ago.

We re-estimate each sector's Pareto tail index to match the top-four firms' sales share in 1992. We use 2-digit NAICS top-four sales shares from 1992 onward, as compiled by (Autor et al., 2020), and extend the series through 2022 using the Establishment and Firm Size Statistics tables. To construct 1992 targets at the 6-digit NAICS level, we assume that the proportional change in the top-four sales share is the same across all industries within each 2-digit NAICS sector.

Conceptually, we attribute the rise in concentration to the emergence of highly productive "superstar firms" (Autor et al., 2020). To focus on the implied change in concentration, rather than any associated changes in the level of brand values, we consider the difference in consumption in the market allocation relative to the perfectly competitive allocation.[15] All other parameters of the model are kept at their 2022 values.

Table 5 summarizes the results. Relative to the perfectly competitive allocation, the welfare gap for all consumers is larger in 2022 than it was in 1992, when market concentration was lower. For high-income consumers, this relative welfare penalty increases by only 0.3 log pts, while for low-income households it increases by 2.4 log pts, 8 times as much. Superstar firms not only charge high prices, they also distort the quality offered to low-income households downwards, increasingly widening the gap to the perfectly competitive allocation over time.

[15] We assume that the scale of the Pareto distribution remains unchanged. This implies that the average brand value in 1992 is lower than in our baseline calibration. Note that the average brand value affects aggregate welfare but not inequality between the rich and the poor.

Table 5: The Distributional Impact of Rising Market Power

	1992		2022	
	High-income	Low-income	High-income	Low-income
Market allocation	+1.9	−4.9	+1.6	−7.3
Island economy	−0.4	−0.4	−1.4	−1.4

Notes: This table reports the log points difference in aggregate consumption of each income type relative to the perfectly competitive allocation in 1992 and 2022. The first row compares the market allocation to the perfectly competitive allocation. The second row compares the island-economy allocation to the perfectly competitive allocation.

We conclude by highlighting that the rise in concentration would not have had a distributional effect absent cannibalization considerations, i.e., in the island economy. The second row of Table 5 shows that in the island economy, the rise in market concentration over this 30-year period would reduce the welfare of all households by 1 log point relative to the perfectly-competitive allocation. This decline in welfare is driven entirely through the misallocation channel, as quality levels are not distorted in the island economy. That is, rising concentration reduces welfare through misallocation in the island economy, but its unequal effects across income groups in the market allocation stem from firms' cannibalization incentives.

6 Conclusion

This paper studies the distributional consequences of market power through the lens of endogenous product design. We show that firms with sufficient market power distort the quality of their low-end products to prevent cannibalization of their high-end offerings, a force that is absent under sufficient competition. Through general equilibrium forces, these distortions benefit high-income households at the expense of low-income ones. Both a reduction in the number of competitors and the emergence of superstar firms amplify consumption inequality through this channel.

Empirical evidence from the NielsenIQ Consumer Panel supports the model's key prediction: in more concentrated markets, dominant firms capture a disproportionately larger share of high-income consumers. In a calibrated version of the model, the distributional costs of market power are large. Market power reduces the welfare of low-income households by 7.3 log points relative to the perfectly-competitive allocation while increasing that of high-income households by 1.6 log points, effects that are several times larger than the standard efficiency costs of imperfect competition. The rise in market concentration since the 1990s widened the welfare gap between rich and poor by 2.1 log points.

Several questions remain open. Our analysis takes market structure as given: the number of firms and their brand values are exogenous. In practice, both may respond to the forces we study. The distributional consequences of market power could feed back into concentration through entry and exit. And if a firm's brand value depends in part on the composition of its customer base, the mechanism we identify may be self-reinforcing: by distorting their low-end products and tilting their customer base toward high-income households, dominant firms may further increase their brand appeal, strengthening the very market power that led to the distortion. Whether promoting competition is more effective at protecting low-income consumers than direct quality regulation, which the historical record suggests is difficult to enforce, is a question we leave for future research.

References

Atkeson, A. and A. Burstein (2008): "Pricing-to-market, trade costs, and international relative prices," *American Economic Review*, 98, 1998–2031. 1, 3.1

Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020): "The Fall of the Labor Share and the Rise of Superstar Firms*," *The Quarterly Journal of Economics*, 135, 645–709. 1, 1, 5.3

Baqaee, D. R. and E. Farhi (2020): "Productivity and misallocation in general equilibrium," *The Quarterly Journal of Economics*, 135, 105–163. 1

Becker, J. (2024): "Do Poor Households Pay Higher Markups in Recessions?" Working paper, New York University. 1

Boar, C. and V. Midrigan (2024): "Markups and Inequality," *Review of Economic Studies*, advance online publication. 1

Bornstein, G. and A. Peter (2025): "Nonlinear Pricing and Misallocation," *American Economic Review*, 115, 3853–3908. 9

De Loecker, J., J. Eeckhout, and G. Unger (2020): "The rise of market power and the macroeconomic implications," *The Quarterly Journal of Economics*, 135, 561–644. 1, 1

Deneckere, R. J. and R. P. McAfee (1996): "Damaged Goods," *Journal of Economics & Management Strategy*, 5, 149–174. 4.1

Dupuit, J. (1849): "De l'influence des péages sur l'utilité des voies de communication," *Annales des Ponts et Chaussées*, 17, 170–248, english translation in *International Economic Papers*, 11:7–31, 1962. 4.1

Edmond, C., V. Midrigan, and D. Y. Xu (2022): "How costly are markups?" *Journal of Political Economy*. 1

European Commission (2009): "Commission Decision of 13 May 2009, Case COMP/C-3/37.990 — Intel," OJ C 227, 22.9.2009, pp. 13–17. 8

Finkelstein, M. O. and R. M. Friedberg (1967): "The application of an entropy theory of concentration to the Clayton Act," *The Yale Law Journal*, 76, 677–717. 2.3

Hsieh, C.-T. and P. J. Klenow (2009): "Misallocation and manufacturing TFP in China and India," *The Quarterly journal of economics*, 124, 1403–1448. 1

Jaravel, X. (2019): "The Unequal Gains from Product Innovations: Evidence from the U.S. Retail Sector," *The Quarterly Journal of Economics*, 134, 715–783. 1

Kaplan, G. and G. Menzio (2015): "The Morphology of Price Dispersion," *International Economic Review*, 56, 1165–1206. 1

LOWE, K. (1993): "Quarterly 80x86 Market Update," Tech. Rep. MCRO-WW-DP-9301, Dataquest, dataquest Perspective: Microcomponents Worldwide. 8

MCLEAN, I. AND C. D. FOSTER (1992): "The Political Economy of Regulation: Interests, Ideology, Voters and the UK Regulation of Railways Act 1844," *Public Administration*, 70, 313–331. 4.1

MIDRIGAN, V. (2011): "Menu costs, multiproduct firms, and aggregate fluctuations," *Econometrica*, 79, 1139–1180. 5.1

MONGEY, S. AND M. E. WAUGH (2025): "Pricing Inequality," Working Paper 33399, NBER. 1

MUSSA, M. AND S. ROSEN (1978): "Monopoly and product quality," *Journal of Economic Theory*, 18, 301–317. 1, 1

NORD, L. (2023): "Shopping, Demand Composition, and Equilibrium Prices," . 1

NVIDIA CORPORATION (2020): "NVIDIA Ampere GA102 GPU Architecture Whitepaper," Technical whitepaper, NVIDIA Corporation, accessed March 2026. 4.1

OBERFIELD, E. (2023): "Inequality and Measured Growth," NBER Working Paper 31096, National Bureau of Economic Research. 1

ROCHET, J.-C. AND L. A. STOLE (2002): "Nonlinear Pricing with Random Participation," *The Review of Economic Studies*, 69, 277–311. 1

SANGANI, K. (2024): "Markups across the income distribution: Measurement and implications," *Available at SSRN 4092068*. 1

STOLE, L. A. (1995): "Nonlinear Pricing and Oligopoly," *Journal of Economics & Management Strategy*, 4, 529–562. 1

WEINTRAUB, S. (2016): "Tesla's New 60kWh Pricing Option Is a Software Revolution," Electrek. 4.1

WOLMAR, C. (2007): *Fire and Steam: A New History of the Railways in Britain*, London: Atlantic Books. 4.1

A Mathematical appendix

PROOF OF LEMMA 1. In equilibrium, each household chooses the product quality that is designated to them. The discrete choice problem of the household is[16]

$$j^* = \arg\max_j (\sigma - 1)\left(\ln(q_{jl}) - \frac{p_{jl}}{P_l} + \eta_j\right) + \epsilon_{ij}. \tag{A.1}$$

Let $\nu_{ji} \equiv (\sigma - 1)(\ln(q_{jl}) - \frac{p_{jl}}{P_l} + \eta_j)$ denote the surplus of consumer i when purchasing their designated bundle from firm j, *net* of the taste shock ϵ_{ij}. That is, ν_{ji} is identical for all low-income (ν_{jl}) and all high-income (ν_{jh}) consumers.

The probability that firm j is chosen, conditional on the taste shock of the household being $\bar{\epsilon}$ is given by

$$\begin{aligned}
Pr\left(j_l^* = j|\epsilon_{ij} = \bar{\epsilon}\right) &= \Pi_{j' \neq j}\left[Pr\left(\nu_{jl} + \bar{\epsilon} \geq \nu_{j'l} + \epsilon_{ij'}\right)\right] \\
&= \Pi_{j' \neq j}\left[Pr\left(\epsilon_{ij'} \leq \nu_{jl} - \nu_{j'l} + \bar{\epsilon}\right)\right] \\
&= \Pi_{j' \neq j} e^{-e^{-\left(\nu_{jl} - \nu_{j'l} + \bar{\epsilon}\right)}} \\
&= \Pi_{j'}\left[e^{-e^{-\left(\nu_{jl} - \nu_{j'l} + \bar{\epsilon}\right)}}\right] e^{e^{-\bar{\epsilon}}} \\
&= e^{-\sum_{j'} e^{-(\nu_{jl} - \nu_{j'l} + \bar{\epsilon})}} e^{e^{-\bar{\epsilon}}} \\
&= e^{-e^{-\bar{\epsilon}} \sum_{j'} e^{-(\nu_{jl} - \nu_{j'l})}} e^{e^{-\bar{\epsilon}}} \\
&= e^{-e^{-\bar{\epsilon}} Q} e^{e^{-\bar{\epsilon}}},
\end{aligned}$$

where $Q = \sum_{j'} e^{-(\nu_{jl} - \nu_{j'l})}$. The market share of firm j for low-income households is derived by integrating over the probability distribution of ϵ_{ij}:

$$\begin{aligned}
B_{jl} &= \int e^{-Qe^{-\bar{\epsilon}}} e^{e^{-\bar{\epsilon}}} e^{-\bar{\epsilon} - e^{-\bar{\epsilon}}} d\bar{\epsilon} \\
&= \int e^{-Qe^{-\bar{\epsilon}}} e^{-\bar{\epsilon}} d\bar{\epsilon} \\
&= \int_0^\infty e^{-Qx} dx \\
&= \frac{1}{Q} \\
&= \frac{e^{\nu_{jl}}}{\sum_{j'} e^{\nu_{j'l}}}.
\end{aligned}$$

where the third equality uses integration by substitution with $x = e^{-\epsilon}$. Thus, we obtained

$$B_{jl} = \frac{e^{\nu_{jl}}}{\sum_{j'} e^{\nu_{j'l}}}. \tag{A.2}$$

[16] We have multiplied the maximand by $\sigma - 1$. This does not change the argmax. It will make the mathematical derivations simpler as will be clear below.

Using the same derivations, we obtain

$$B_{jh} = \frac{e^{\nu_{jh}}}{\sum_{j'} e^{\nu_{j'h}}}. \tag{A.3}$$

That is, we have proved that

$$\mathcal{B}\left(q_j, p_j, \eta_j; \{q_{j'}, p_{j'}, \eta_{j'}\}_{j' \neq j}, P\right) = \frac{e^{(\sigma-1)\left(\ln(q_j) - \frac{p_j}{P} + \eta_j\right)}}{\sum_{j'} e^{(\sigma-1)\left(\ln(q_{j'}) - \frac{p_{j'}}{P} + \eta_{j'}\right)}}. \tag{A.4}$$

■

Proof of Proposition 1. As in the proof of Lemma 1, in equilibrium each household purchases the bundle designated to their type, so choosing firm j in a sector yields the systematic payoff $\ln q_{j\tau} + \eta_j - p_{j\tau}/P_\tau$ plus the idiosyncratic taste $\varepsilon_{ij}/(\sigma-1)$. The proof proceeds in three steps.

Step 1: Separation. Since menus are identical across sectors, the price paid and the systematic payoff depend only on *which* firm is chosen, not on the sector. Hence total expenditure and the first term of utility depend on the household's sector-to-firm assignment only through the fractions $B_{j\tau} = |A_j|$, where A_j is the set of sectors in which firm j is chosen. The assignment affects utility only through the taste term. The household problem (2.3) therefore separates into an outer choice of fractions and an inner assignment problem:

$$\ln C_\tau = \max_{\{B_{j\tau}\}} \sum_j B_{j\tau} \left(\ln q_{j\tau} + \eta_j\right) + \frac{V(\{B_{j\tau}\})}{\sigma - 1} \quad \text{s.t.} \quad \sum_j B_{j\tau}\, p_{j\tau} = E_\tau, \quad \sum_j B_{j\tau} = 1, \quad B_{j\tau} \geq 0,$$

where

$$V(\{B_{j\tau}\}) \equiv \sup_{\{A_j\}:\, |A_j| = B_{j\tau}} \sum_j \int_{A_j} \varepsilon_{ij}(s)\, ds.$$

By the law of large numbers across the continuum of sectors, $V(\{B_{j\tau}\})$ is deterministic and common to all households of a given type.

Step 2: The assignment value is the entropy measure. We show $V(\{B_{j\tau}\}) = \gamma + \sum_j B_{j\tau} \ln(1/B_{j\tau})$.

Upper bound. For any constants $\{a_j\}$ and any admissible partition $\{A_j\}$, since $\varepsilon_{ij}(s) \leq \max_{j'} \left(\varepsilon_{ij'}(s) + a_{j'}\right) - a_j$ for $s \in A_j$,

$$\sum_j \int_{A_j} \varepsilon_{ij}(s)\, ds \;\leq\; \int \max_{j'} \left(\varepsilon_{ij'}(s) + a_{j'}\right) ds - \sum_j a_j B_{j\tau} \;=\; \ln \sum_{j'} e^{a_{j'}} + \gamma - \sum_j a_j B_{j\tau},$$

where the equality uses the law of large numbers across the continuum of sectors and the expected maximum of Gumbel random variables. Setting $a_j = \ln B_{j\tau}$, the bound becomes $\gamma + \sum_j B_{j\tau} \ln(1/B_{j\tau})$, since $\ln \sum_j e^{\ln B_{j\tau}} = \ln 1 = 0$.

Attainment. The bound is attained by a threshold rule in which each sector is assigned to the firm with the highest taste draw, after handicapping each firm j by the constant $\ln B_{j\tau}$, so that firm j is chosen in exactly a fraction $B_{j\tau}$ of sectors. Formally, consider the assignment

$$A_j = \left\{ s : j = \arg\max_{j'} \left(\epsilon_{ij'}(s) + \ln B_{j'\tau} \right) \right\}.$$

We first verify admissibility. By the derivation in the proof of Lemma 1 (with $\nu_{j'} = \ln B_{j'\tau}$), the probability that firm j attains the maximum, conditional on the taste draw for firm j being $\bar{\epsilon}$, is

$$Pr\left(j^* = j \,|\, \epsilon_{ij} = \bar{\epsilon}\right) = e^{-Q_j e^{-\bar{\epsilon}}} e^{e^{-\bar{\epsilon}}}, \qquad \text{where} \quad Q_j = \sum_{j'} e^{-\left(\ln B_{j\tau} - \ln B_{j'\tau}\right)} = \frac{1}{B_{j\tau}} \sum_{j'} B_{j'\tau} = \frac{1}{B_{j\tau}},$$

and, integrating over the distribution of $\bar{\epsilon}$, firm j is chosen with probability $1/Q_j = B_{j\tau}$. By the law of large numbers across the continuum of sectors, the measure of A_j is $B_{j\tau}$, so the assignment is admissible.

Next, we compute the taste shocks harvested on A_j. Using the distribution of ϵ_{ij},

$$\begin{aligned}
\int_{A_j} \epsilon_{ij}(s)\, ds &= \int e^{-Q_j e^{-\bar{\epsilon}}} e^{e^{-\bar{\epsilon}}} e^{-\bar{\epsilon} - e^{-\bar{\epsilon}}} \bar{\epsilon}\, d\bar{\epsilon} \\
&= \int e^{-Q_j e^{-\bar{\epsilon}}} e^{-\bar{\epsilon}} \bar{\epsilon}\, d\bar{\epsilon} \\
&= -\int_0^\infty e^{-Q_j x} \ln(x)\, dx \\
&= \frac{\gamma + \ln Q_j}{Q_j} = B_{j\tau}\left(\gamma + \ln \frac{1}{B_{j\tau}}\right),
\end{aligned}$$

where the first equality uses the law of large numbers across sectors, the third equality uses integration by substitution with $x = e^{-\bar{\epsilon}}$, and γ is the Euler–Mascheroni constant. Summing across firms, the assignment achieves

$$\sum_j \int_{A_j} \epsilon_{ij}(s)\, ds = \sum_j B_{j\tau}\left(\gamma + \ln \frac{1}{B_{j\tau}}\right) = \gamma + \sum_j B_{j\tau} \ln \frac{1}{B_{j\tau}},$$

which coincides with the upper bound. Note that the tilts $\ln B_{j\tau}$ act as the shadow prices of the measure constraints: the marginal sector assigned to firm j is indifferent between j and its best alternative, so the foregone taste draws are fully priced in.

Step 3: The outer problem. Substituting $V(\{B_{j\tau}\})$ yields (2.11). The objective is the sum of a linear function and the strictly concave entropy, so the problem has a unique solution, which is interior since the entropy has unbounded slope as $B_{j\tau} \to 0$ (Inada condition). ■

Proof of Proposition **2.**

The proof proceeds in three steps: We start by showing that, in any perfectly competitive allocation, we must have

1. $p_{j\tau}^{PC} = q_{j\tau}^{PC}\kappa$,
2. $q_{j\tau}^{PC} = \frac{P_\tau^{PC}}{\kappa}$, where P_τ^{PC} is the equilibrium marginal price index,
3. and, in equilibrium, $P_\tau^{PC} = L_\tau$.

Last, we show that (2.13) and (2.14) directly follow.

Since all potential producers of variety j share η_j as well as the idiosyncratic draws of ε_{ij}, all consumers will purchase from the producer who offers maximal surplus, given by $\ln(q) - \frac{p}{P}$.

1. Suppose that $p_{j\tau}^{PC} \neq q_{j\tau}^{PC}\kappa$. If $p_{j\tau}^{PC} < q_{j\tau}^{PC}\kappa$, any firm makes negative profits and would exit. Suppose instead $p_{j\tau}^{PC} > q_{j\tau}^{PC}\kappa$, and the firm offering that price and quantity commands a market share $B_{j\tau}$. Then, a new entrant could offer $q_{j\tau}^{PC}$ at $p_{j\tau}^{'} = p_{j\tau}^{PC} - \delta$ for some $\delta > 0$. Their market share $B_{j\tau}^{'}$ would be at least as large as $B_{j\tau}$, implying positive profits for the potential entrant. Hence, only $p_{j\tau}^{PC} = q_{j\tau}^{PC}\kappa$ is an equilibrium.

2. Suppose that $q_{j\tau}^{PC} > (<)\frac{P_\tau^{PC}}{\kappa}$. Note that, since $p = \kappa q$ from part (1) above, consumer surplus, $\ln(q) - \frac{q\kappa}{P}$ is maximized at $q = \frac{P}{\kappa}$. Then, there exists $q' < (>)q_{j\tau}^{PC}$, such that, at $p' = q'\kappa$, firms make zero profits and consumers strictly prefer $\{q', p'\}$. Zero profits follows directly from the linearity of the firm's profit function. Firms could therefore offer a bundle $\{q', p' - \delta\}$ for some $\delta > 0$ such that consumers are indifferent and make positive profits.

3. Since firms make zero profits, the household budget constraints imply that total expenditure equals labor income, $E_\tau = L_\tau$. By parts (1) and (2), every firm charges the same price in segment τ: $p_{j\tau}^{PC} = \kappa q_{j\tau}^{PC} = P_\tau^{PC}$. Because the household purchases one product in each of a measure one of sectors, total expenditure is $E_\tau = \int p_{j(s)\tau}^{PC}\, ds = P_\tau^{PC}$. Hence $P_\tau^{PC} = L_\tau$.

Note that since equilibrium allocations in the perfectly competitive environment maximize consumer surplus, incentive compatibility constraints are slack.

Using $P_\tau^{PC} = L_\tau$, condition (2.13) follows directly from $q_{j\tau}^{PC} = \frac{P_\tau^{PC}}{\kappa}$, and condition (2.14) follows from the definition of the market share in (2.6). ■

Proof of Lemma **2.**

Using Proposition 1 and the fact that $B_{j\tau}$ are identical on the two consumer segments and $q_{j\tau}$ are constant across firms within a consumer segment by Proposition 2, consumption inequality reduces to $\ln(q_h) - \ln(q_l)$. Plugging in equilibrium quantities from Proposition 2 gives the expressions in Lemma 2. ■

Proof of Proposition **3.** Taking first-order conditions of the firm's profit maximization problem in equation (3.1), we obtain

$$[p_{j\tau}]: \qquad B_{j\tau} + (p_{j\tau} - \kappa q_{j\tau}) \frac{\partial B_{j\tau}}{\partial p_{j\tau}} = 0,$$

$$[q_{j\tau}]: \qquad -\kappa B_{j\tau} + (p_{j\tau} - \kappa q_{j\tau}) \frac{\partial B_{j\tau}}{\partial q_{j\tau}} = 0.$$

Differentiating $B_{j\tau}$ and combining the two equations, we get the following optimality conditions for prices and quantities:

$$\frac{1}{q_{j\tau}} = \frac{\kappa}{P_\tau} \tag{A.5}$$

$$p_{j\tau} = \kappa q_{j\tau} + \frac{P_\tau}{(\sigma-1)(1-B_{j\tau})} \tag{A.6}$$

Given the firms' optimality conditions above, it remains to prove that $P_\tau^{IE} = L_\tau$ is the unique equilibrium. Note that from Equations (A.5) & (A.6) it already follows that $q_{j\tau} = q_\tau$ is constant across firms j and $B_{j\tau} = B_j$ is constant across consumer segments.

Using equations (A.5) & (A.6) and the definition of firm profits from (2.5), aggregate profits simplify to

$$\Pi = \frac{P_h + P_l}{2} \sum_j \frac{B_j}{(\sigma-1)(1-B_j)}. \tag{A.7}$$

Labor market clearing simplifies to

$$\frac{\kappa}{2}(q_l + q_h) = 1, \tag{A.8}$$

which, using (A.5), implies that $P_h + P_l = 2$.

Given the above simplifications, as well as the fact that $L_l = 1 - \alpha$, the budget constraint for low-income households becomes:

$$\sum_j B_j p_{jl} = (1-\alpha)\Pi + L_l \tag{A.9}$$

$$\sum_j B_j P_l \left(1 + \frac{1}{(\sigma-1)(1-B_j)}\right) = (1-\alpha) \sum_j \frac{B_j}{(\sigma-1)(1-B_j)} + 1 - \alpha \tag{A.10}$$

$$P_l \left(1 + \sum_j \frac{B_j}{(\sigma-1)(1-B_j)}\right) = (1-\alpha) \left(1 + \sum_j \frac{B_j}{(\sigma-1)(1-B_j)}\right). \tag{A.11}$$

Since $B_j \in (0,1)$, $\left(1 + \sum_j \frac{B_j}{(\sigma-1)(1-B_j)}\right) \neq 0$ and hence, $P_l = (1-\alpha)$ is the unique solution to (A.11). An analogous argument with the high-income budget constraint proves that $P_h = (1+\alpha)$.

It remains to show that the market shares are uniquely determined by (3.5). Define $\phi(B) \equiv$

$\ln B + \frac{1}{1-B}$ on $(0,1)$. Taking logs of (3.5), any solution satisfies

$$\phi(B_j) = (\sigma - 1)\eta_j - c, \qquad c \equiv \ln \sum_{j'} e^{(\sigma-1)\eta_{j'} - \frac{1}{1-B_{j'}}}, \tag{A.12}$$

together with $\sum_j B_j = 1$; conversely, any $\{B_j\}$ satisfying (A.12) for some constant c and summing to one solves (3.5). Since $\phi'(B) = \frac{1}{B} + \frac{1}{(1-B)^2} > 0$, the function ϕ is strictly increasing. Therefore, for every c, the candidate shares $B_j(c) = \phi^{-1}((\sigma-1)\eta_j - c)$ are uniquely determined, and each is continuous and strictly decreasing in c, with $B_j(c) \to 1$ as $c \to -\infty$ and $B_j(c) \to 0$ as $c \to \infty$. The sum $\sum_j B_j(c)$ therefore decreases continuously and strictly from N to 0, so there is exactly one c^* at which it equals one. The market shares $B_j = B_j(c^*)$ are therefore unique. Note that this argument also establishes existence.

■

Proof of Lemma **3.**

Fix a consumer type $\tau \in \{l, h\}$. Taking the ratio of the island-economy market shares of firms j and k in equation (3.5), we obtain

$$\frac{B^{IE}_{j\tau}}{B^{IE}_{k\tau}} = \exp\left((\sigma - 1)(\eta_j - \eta_k) - \left[\frac{1}{1 - B^{IE}_{j\tau}} - \frac{1}{1 - B^{IE}_{k\tau}}\right]\right). \tag{A.13}$$

We first show that $\eta_j > \eta_k$ implies $B^{IE}_{j\tau} > B^{IE}_{k\tau}$. Suppose instead that $B^{IE}_{j\tau} \leq B^{IE}_{k\tau}$. Since $1/(1-B)$ is strictly increasing in B,

$$\frac{1}{1 - B^{IE}_{j\tau}} - \frac{1}{1 - B^{IE}_{k\tau}} \leq 0. \tag{A.14}$$

Since $\eta_j > \eta_k$ and $\sigma > 1$, the expression on the right-hand side of the market-share ratio is strictly larger than one. This contradicts $B^{IE}_{j\tau}/B^{IE}_{k\tau} \leq 1$. Hence, $B^{IE}_{j\tau} > B^{IE}_{k\tau}$.

It follows that

$$\frac{1}{1 - B^{IE}_{j\tau}} - \frac{1}{1 - B^{IE}_{k\tau}} > 0, \tag{A.15}$$

and therefore

$$\frac{B^{IE}_{j\tau}}{B^{IE}_{k\tau}} < \exp\left((\sigma - 1)(\eta_j - \eta_k)\right). \tag{A.16}$$

Using the perfectly competitive market shares in equation (2.14),

$$\exp\left((\sigma - 1)(\eta_j - \eta_k)\right) = \frac{B^{PC}_{j\tau}}{B^{PC}_{k\tau}}. \tag{A.17}$$

Thus,

$$\frac{B^{IE}_{j\tau}}{B^{IE}_{k\tau}} < \frac{B^{PC}_{j\tau}}{B^{PC}_{k\tau}}, \tag{A.18}$$

which directly implies

$$\frac{B^{IE}_{j\tau}}{B^{PC}_{j\tau}} < \frac{B^{IE}_{k\tau}}{B^{PC}_{k\tau}}. \tag{A.19}$$

■

Proof of Proposition 4.

We start by proving the first part, $\ln C^{IE}_{\tau} \leq \ln C^{PC}_{\tau}$. From Propositions (2) and (3), we know that quality levels in the island economy are identical to those in the perfectly competitive allocation. Thus, the two allocations differ only in their market shares.

Using Proposition 1 and the fact that $q_{j\tau} = q_{\tau}$ is constant across firms, consumption of households of type τ can be written as

$$\ln C_{\tau} = \ln q_{\tau} + \sum_j B_{j\tau}\eta_j + \frac{1}{\sigma-1}\sum_j B_{j\tau}\ln(1/B_{j\tau}) + \frac{\gamma}{\sigma-1}. \tag{A.20}$$

The terms depending on market shares are strictly concave in $\{B_{j\tau}\}_j$. Maximizing them subject to $\sum_j B_{j\tau} = 1$ gives

$$B_{j\tau} = \frac{e^{(\sigma-1)\eta_j}}{\sum_{j'} e^{(\sigma-1)\eta_{j'}}} = B^{PC}_{j\tau}. \tag{A.21}$$

Hence, the perfectly competitive market shares maximize consumption conditional on q_{τ}. Since $q^{IE}_{\tau} = q^{PC}_{\tau}$, it follows that

$$\ln C^{IE}_{\tau} \leq \ln C^{PC}_{\tau}, \qquad \forall \tau \in \{l,h\}. \tag{A.22}$$

The inequality is strict whenever market shares are not identical to the PC allocation, which is the case for the island economy by Lemma 3 as long as brand values are not identical across all firms.

Part 2 of the lemma, $\ln C^{IE}_{h} - \ln C^{IE}_{l} = \ln C^{PC}_{h} - \ln C^{PC}_{l}$ follows directly from (i) $q^{IE}_{j\tau} = q^{PC}_{\tau} \equiv q_{\tau}$ for all firms j and $B^{IE}_{j\tau} = B^{IE}_{j}$ is identical in both consumer segments. Using these, we can write aggregate consumption fo type τ consumers in both the island economy allocation ($x = IE$) as well as the perfectly competitive ($x = PC$) one as:

$$\ln C^{x}_{\tau} = \ln q_{\tau} + \sum_j B^{x}_{j}\eta_j + \frac{\sum_j B^{x}_{j}\ln(1/B^{x}_{j})}{\sigma-1} + \frac{\gamma}{\sigma-1}. \tag{A.23}$$

Taking differences between high- and low-income consumers, all terms including market shares cancel and we are left with the difference in the quality level between the two consumer segments.

$$\ln C_h^x - \ln C_l^x = \ln q_h - \ln q_l. \tag{A.24}$$

■

Proof of Lemma 4. Using Proposition 3 and the firms' optimality conditions, quantities and prices in the island economy simplify to

$$q_{j\tau}^{IE} = \frac{L_\tau}{\kappa}, \qquad p_{j\tau}^{IE} = L_\tau \left(1 + \frac{1}{(\sigma-1)(1-B_j^{IE})}\right), \qquad \tau \in \{l,h\}. \tag{A.25}$$

Define the brand-specific markup as

$$\widetilde{\mu}_j \equiv 1 + \frac{1}{(\sigma-1)(1-B_j^{IE})} > 1, \tag{A.26}$$

so that $p_{j\tau}^{IE} = L_\tau \widetilde{\mu}_j$.

First, we consider the incentive compatibility constraint of low-income households. Substituting the island-economy allocation into the low-income IC constraint gives

$$\ln\left(\frac{L_h}{L_l}\right) \leq \frac{p_{jh}^{IE} - p_{jl}^{IE}}{P_l} = \widetilde{\mu}_j \left(\frac{L_h}{L_l} - 1\right). \tag{A.27}$$

Since $\ln(x) < x - 1$ for $x > 1$ and $\widetilde{\mu}_j > 1$, we have

$$\ln\left(\frac{L_h}{L_l}\right) < \frac{L_h}{L_l} - 1 < \widetilde{\mu}_j \left(\frac{L_h}{L_l} - 1\right). \tag{A.28}$$

Hence, the incentive compatibility constraint of low-income households is strictly slack for every firm.

Next, we consider the incentive compatibility constraint of high-income households. Substituting the island-economy allocation gives

$$\ln\left(\frac{L_h}{L_l}\right) \geq \frac{p_{jh}^{IE} - p_{jl}^{IE}}{P_h} = \widetilde{\mu}_j \left(1 - \frac{L_l}{L_h}\right). \tag{A.29}$$

Thus, the high-income IC constraint is violated whenever

$$\widetilde{\mu}_j > \frac{\ln(L_h/L_l)}{1 - L_l/L_h}. \tag{A.30}$$

The threshold on the right-hand side is independent of the firm. Moreover,

$$\frac{\partial \widetilde{\mu}_j}{\partial B_j^{IE}} = \frac{1}{(\sigma-1)(1-B_j^{IE})^2} > 0. \tag{A.31}$$

As established in the proof of Lemma 3, B_j^{IE} is increasing in the firm's brand value η_j. It follows that $\widetilde{\mu}_j$ is also increasing in η_j. Therefore, if the high-income IC constraint is violated for a firm with brand value $\bar{\eta}$, it is violated for every firm with $\eta_j \geq \bar{\eta}$. Ordering firms by their brand values, the set of firms for which the high-income IC constraint is violated therefore consists of the top $\widetilde{N}$ brands. ■

Proof of Proposition 5.

To derive the optimal product portfolio of the firm in the decentralized equilibrium, we proceed in two steps. First, we characterize the unconstrained problem where the incentive compatibility is slack. Second, if the unconstrained allocation violates the incentive compatibility constraint for the high-type, we solve the combined problem while imposing that the incentive compatibility constraint holds with equality.

Unconstrained case. When the incentive compatibility is slack, i.e., $\psi = 0$, the firm's problem is separable across the two consumer types and is identical to the firm's problem in the *Island Economy*. Using the results from Proposition 3, we have

$$q_{j\tau} = \frac{P_\tau}{\kappa}, \tag{A.32}$$

$$p_{j\tau} = \kappa q_{j\tau} \tilde{\mu}_{j\tau}, \tag{A.33}$$

where $\tilde{\mu}_{j\tau} \equiv \left(\frac{\sigma}{\sigma-1} + \frac{1}{\sigma-1} \frac{B_{j\tau}}{1-B_{j\tau}} \right)$.

Constrained allocation. Suppose that the IC-high constraint is binding, and let us characterize the constrained allocation. The firm's problem is given by

$$\begin{aligned} \max_{\{q_l, q_h, p_l, p_h, B_l, B_h\}} \quad & B_l \left(p_l - \kappa q_l\right) + B_h \left(p_h - \kappa q_h\right), \\ \text{s.t.} \quad & B_l = \frac{e^{(\sigma-1)\left(\ln(q_l) - \frac{p_l}{P_l} + \eta_j\right)}}{e^{(\sigma-1)\left(\ln(q_l) - \frac{p_l}{P_l} + \eta_j\right)} + S_{-jl}}, \\ & B_h = \frac{e^{(\sigma-1)\left(\ln(q_h) - \frac{p_h}{P_h} + \eta_j\right)}}{e^{(\sigma-1)\left(\ln(q_h) - \frac{p_h}{P_h} + \eta_j\right)} + S_{-jh}}, \\ & \ln(q_h) - \ln(q_l) = \frac{p_h - p_l}{P_h} \end{aligned} \tag{A.34}$$

Taking FOC:

$$\begin{aligned}
[B_l]: \qquad & p_l - \kappa q_l = \lambda_l, \\
[B_h]: \qquad & p_h - \kappa q_h = \lambda_h, \\
[q_l]: \qquad & \kappa q_l = \lambda_l(\sigma-1)\left(1-B_l\right) - \psi\frac{1}{B_l}, \\
[q_h]: \qquad & \kappa q_h = \lambda_h(\sigma-1)\left(1-B_h\right) + \psi\frac{1}{B_h}, \\
[p_l]: \qquad & 1 = \lambda_l(\sigma-1)\left(1-B_l\right)\frac{1}{P_l} - \psi\frac{1}{P_h B_l}, \\
[p_h]: \qquad & 1 = \lambda_h(\sigma-1)\left(1-B_h\right)\frac{1}{P_h} + \psi\frac{1}{P_h B_h},
\end{aligned}$$

where λ_h and λ_l are the Lagrange multipliers on the market share demand functions and ψ is the Lagrange multiplier on the IC constraint. Combining the FOCs with respect to q_h and p_h, we get that the quantity sold to the high type equates marginal utility to marginal cost

$$q_h = \frac{P_h}{\kappa}. \tag{A.35}$$

So that $\lambda_h = p_h - P_h$ from the second optimality condition. Combining the the optimality conditions w.r.t. q_l and p_l, we obtain

$$q_l = \frac{P_l}{\kappa} - \frac{\psi}{\kappa B_l}\left(1 - \frac{P_l}{P_h}\right). \tag{A.36}$$

The equation above shows that the IC constraint leads to a downward distortion in the low-income quality bundle ($q_l < \frac{P_l}{\kappa}$). The optimality conditions w.r.t. q_h and B_h yield

$$\psi = B_h P_h - (p_h - P_h)(\sigma-1)B_h(1-B_h). \tag{A.37}$$

Rearranging to isolate p_h, we have

$$p_h = P_h\left(1 + \frac{1}{\sigma-1}\frac{1}{1-B_h}\right) - \frac{1}{\sigma-1}\frac{1}{B_h(1-B_h)}\psi. \tag{A.38}$$

Using $P_h = \kappa q_h$ from optimal quality choice and the identity $\tilde{\mu}_h = \left(1 + \frac{1}{\sigma-1}\frac{1}{1-B_h}\right)$, we have

$$p_h = \kappa q_h \tilde{\mu}_h - \frac{\psi}{B_h}\left(\tilde{\mu}_h - 1\right). \tag{A.39}$$

Similarly, the optimality conditions w.r.t. q_l and B_l imply:

$$\kappa q_l = (p_l - \kappa q_l)(\sigma-1)\left(1-B_l\right) - \psi\frac{1}{B_l}. \tag{A.40}$$

Isolating p_l we obtain

$$p_l = \kappa q_l \tilde{\mu}_l + \frac{\psi}{B_l}\left(\tilde{\mu}_l - 1\right). \tag{A.41}$$

Finally, the proof is completed by noticing that when $\psi = 0$ equations (A.35), (A.36), (A.39), and (A.41) are identical to their corresponding levels in the unconstrained case. ■

Proof of Proposition **6.** We follow the regular equilibrium branch as $\Delta \downarrow 0$. As in Proposition 5, we assume that the low-income incentive constraints remain slack. Define

$$\mathcal{I}_N \equiv \lim_{\Delta \downarrow 0} \left[\ln C_h(N, \Delta) - \ln C_l(N, \Delta)\right].$$

By continuity and symmetry, in the limit all firms have the same market share, price, quality, and multiplier within each consumer segment. In particular, $B_{j\tau} = 1/N$ for all firms and both consumer types. The argument below also shows that this limiting allocation is unique.

The proof proceeds in three steps:

1. If the high-income IC is binding, consumption inequality is higher than income inequality and decreasing in N.

2. If the high-income IC is slack, consumption inequality is equal to income inequality.

3. If the high-income IC is slack for N, it is slack for all $N' > N$.

Part 1. Suppose first that the high-income IC is binding. Since all firms are identical in the limit, they all sell the same quantity q_τ for the same price p_τ to each consumer type $\tau \in \{l, h\}$. The household budget constraints are therefore

$$p_h = (1 + \alpha)(1 + \Pi), \tag{A.42}$$

$$p_l = (1 - \alpha)(1 + \Pi). \tag{A.43}$$

Because each household type has measure one-half, the resource constraint is

$$\kappa(q_l + q_h) = 2. \tag{A.44}$$

Let

$$\widetilde{\sigma}_N \equiv (\sigma - 1)\left(1 - \frac{1}{N}\right).$$

Using Proposition 5 and $B_{j\tau} = 1/N$, the firm first-order conditions can be written as

$$q_l = \frac{\widetilde{\sigma}_N p_l - N\psi}{\kappa(1 + \widetilde{\sigma}_N)}, \tag{A.45}$$

$$q_h = \frac{\widetilde{\sigma}_N p_h + N\psi}{\kappa(1 + \widetilde{\sigma}_N)}. \tag{A.46}$$

Adding these equations and using (A.44) gives

$$p_h + p_l = 2\left(1 + \frac{1}{\widetilde{\sigma}_N}\right).$$

Hence aggregate profit per household is

$$\Pi = \frac{1}{\widetilde{\sigma}_N},$$

and the prices paid by the two consumer types are

$$p_h = (1+\alpha)\left(1 + \frac{1}{\widetilde{\sigma}_N}\right), \tag{A.47}$$

$$p_l = (1-\alpha)\left(1 + \frac{1}{\widetilde{\sigma}_N}\right). \tag{A.48}$$

We now show that consumption inequality is decreasing in N. Let

$$L \equiv \ln \frac{q_h}{q_l}, \qquad s(L) \equiv \frac{e^L}{1+e^L}.$$

Since brand values, market shares, and entropy terms are the same for rich and poor households in the equal-brand limit, L is consumption inequality. Moreover,

$$s(L) = \frac{q_h}{q_l + q_h}.$$

Using (A.44) and the optimality condition $q_h = P_h/\kappa$ gives

$$\kappa q_h = 2s(L), \tag{A.49}$$

$$P_h = 2s(L). \tag{A.50}$$

Since the high-income IC binds,

$$L = \frac{p_h - p_l}{P_h} \tag{A.51}$$

$$= \frac{2\alpha\,(1 + 1/\widetilde{\sigma}_N)}{2s(L)}. \tag{A.52}$$

Therefore

$$s(L)L = \alpha\left(1 + \frac{1}{(\sigma-1)(1-1/N)}\right). \tag{A.53}$$

The left-hand side is strictly increasing in L, since

$$\frac{d}{dL}[s(L)L] = s(L) + Ls(L)[1-s(L)] > 0.$$

The right-hand side of (A.53) is strictly decreasing in N. Hence L, and therefore consumption inequality, is strictly decreasing in N whenever the high-income IC binds.

It is also useful to note that the solution to (A.53) is unique. At the value of N for which the IC is just binding, it satisfies $L = \ln[(1+\alpha)/(1-\alpha)]$. Thus throughout the binding region,

$$L > \ln\left(\frac{1+\alpha}{1-\alpha}\right),$$

so consumption inequality is strictly larger than income inequality.

Part 2. Suppose instead that the high-income IC is slack. The limiting allocation is then the same as the island allocation. In particular,

$$q_h = \frac{P_h}{\kappa}, \qquad q_l = \frac{P_l}{\kappa},$$

and the normalized firm problem is identical for the two consumer types. General equilibrium therefore gives

$$\frac{P_l}{P_h} = \frac{1-\alpha}{1+\alpha}.$$

Hence

$$\mathcal{I}_N = \ln \frac{q_h}{q_l} = \ln\left(\frac{1+\alpha}{1-\alpha}\right),$$

so consumption inequality equals income inequality and is independent of N.

Part 3. It remains to show that once the high-income IC is slack, adding firms cannot make it bind again. Using $B_{j\tau} = 1/N$, the high-income IC evaluated at the unconstrained allocation can be written as

$$\frac{-\ln(P_l/P_h)}{1-P_l/P_h} \geq 1 + \frac{1}{\sigma-1}\frac{N}{N-1}. \tag{A.54}$$

When the IC is slack, Part 2 gives $P_l/P_h = (1-\alpha)/(1+\alpha)$, so the left-hand side of (A.54) is independent of N. The right-hand side is strictly decreasing in N. Therefore, if the IC is slack for N, it remains slack for every $N' > N$.

Combining the three parts, if screening is active at N, increasing N either leaves the IC binding, in which case Part 1 implies that inequality strictly falls, or eventually makes the IC slack, in which case inequality reaches the income-inequality level from above. Once the IC is slack, Parts 2 and 3 imply that inequality remains constant. Hence

$$\mathcal{I}_{N+1} \leq \mathcal{I}_N$$

for every $N \geq 2$, with strict inequality whenever high-income ICs are binding at N. ■

Proof of Proposition **7.** For every $\Delta > 0$, consider the connected interior equilibrium branch that converges to a symmetric allocation as $\Delta \downarrow 0$. As in Proposition 5, low-income incentive constraints remain slack. We allow the high-income incentive constraints to be either slack or binding in the equal-brand limit. We assume that the equilibrium branch is piecewise continuously differentiable and that changes in the set of binding constraints are regular.

We use two implications of Proposition 5 repeatedly. First,

$$q_{jh} = \frac{P_h}{\kappa}, \tag{A.55}$$

$$q_{jl} = \frac{P_l}{\kappa} - \frac{\psi_j}{\kappa B_{jl}}\left(1 - \frac{P_l}{P_h}\right), \tag{A.56}$$

where $\psi_j = 0$ when firm j's high-income IC is slack. Second, the price first-order conditions imply

$$\frac{p_{jh}}{P_h} = 1 + \frac{1}{(\sigma-1)(1-B_{jh})} - \frac{\psi_j/P_h}{(\sigma-1)B_{jh}(1-B_{jh})}, \tag{A.57}$$

$$\frac{p_{jl}}{P_l} = \frac{\kappa q_{jl}}{P_l} + \frac{1 - \frac{P_l}{P_h}\frac{\kappa q_{jl}}{P_l}}{(\sigma-1)\left(1-\frac{P_l}{P_h}\right)(1-B_{jl})}. \tag{A.58}$$

The market-share equations can be written as

$$\frac{1}{\sigma-1}\ln\frac{B_{2h}}{1-B_{2h}} = \Delta - \frac{p_{2h}-p_{1h}}{P_h}, \tag{A.59}$$

$$\frac{1}{\sigma-1}\ln\frac{B_{2l}}{1-B_{2l}} = \Delta + \ln\frac{q_{2l}}{q_{1l}} - \frac{p_{2l}-p_{1l}}{P_l}. \tag{A.60}$$

We will also use average expenditure normalized by the corresponding price index,

$$X \equiv B_{2h}\frac{p_{2h}}{P_h} + (1-B_{2h})\frac{p_{1h}}{P_h}, \tag{A.61}$$

$$Y \equiv B_{2l}\frac{p_{2l}}{P_l} + (1-B_{2l})\frac{p_{1l}}{P_l}. \tag{A.62}$$

These two definitions save substantial notation below. Since profits are rebated in proportion to labor endowments, $E_h/E_l = (1+\alpha)/(1-\alpha)$. The household budget constraints therefore imply

$$X = \frac{1+\alpha}{1-\alpha}\frac{P_l}{P_h}Y. \tag{A.63}$$

Proposition 1 and the high-income market-share equation imply the two identities

$$\begin{aligned}\ln C_h - \ln C_l = {}& \alpha\left(X + \frac{P_l}{P_h}Y\right) + B_{2l}\left[-\ln\frac{\kappa q_{2l}}{P_h} - \left(\frac{p_{2h}}{P_h} - \frac{p_{2l}}{P_h}\right)\right] \\ &+ (1-B_{2l})\left[-\ln\frac{\kappa q_{1l}}{P_h} - \left(\frac{p_{1h}}{P_h} - \frac{p_{1l}}{P_h}\right)\right] \\ &+ \frac{1}{\sigma-1}D_{\mathrm{KL}}(B_{2l}\|B_{2h}),\end{aligned} \tag{A.64}$$

where

$$D_{\mathrm{KL}}(B_{2l}\|B_{2h}) \equiv B_{2l}\ln\frac{B_{2l}}{B_{2h}} + (1-B_{2l})\ln\frac{1-B_{2l}}{1-B_{2h}} \geq 0,$$

and

$$\begin{aligned}\ln C_h - \ln C_l = & - B_{2l} \ln \frac{\kappa q_{2l}}{P_h} - (1 - B_{2l}) \ln \frac{\kappa q_{1l}}{P_h} \\ & + (B_{2h} - B_{2l}) \left(\frac{p_{2h} - p_{1h}}{P_h} \right) + \frac{1}{\sigma - 1} D_{\mathrm{KL}}(B_{2l} \| B_{2h}). \end{aligned} \tag{A.65}$$

The two bracketed terms in (A.64) are exactly the slacks in the two high-income IC constraints, and are therefore nonnegative.

No screening. Suppose first that both high-income IC constraints are slack. Then $\psi_1 = \psi_2 = 0$, $q_{jl} = P_l/\kappa$, and the normalized firm problem is identical for rich and poor households. Hence

$$B_{2h} = B_{2l}.$$

Average normalized expenditure is also the same for the two household types, so (A.63) gives

$$\frac{P_l}{P_h} = \frac{1 - \alpha}{1 + \alpha}.$$

Proposition 1 then implies

$$\ln C_h - \ln C_l = \ln \left(\frac{1 + \alpha}{1 - \alpha} \right). \tag{A.66}$$

Thus brand dispersion has no effect on consumption inequality as long as neither firm screens.

At the equal-brand limit, each firm has market share 1/2, so the unconstrained normalized price is $1 + 2/(\sigma - 1)$. The unconstrained limit is feasible if and only if

$$1 + \frac{2}{\sigma - 1} \leq \frac{-\ln\left((1 - \alpha)/(1 + \alpha)\right)}{1 - (1 - \alpha)/(1 + \alpha)}. \tag{A.67}$$

If this inequality is strict, both high-income IC constraints remain slack for an initial interval of Δ. Over that interval, $B_{2h} = B_{2l}$ rises with Δ, so firm 2's desired price rises while firm 1's falls. Firm 2 is therefore the first firm whose high-income IC can bind.

Orderings once screening is active. Suppose first that only firm 2's high-income IC binds. Substituting (A.55)–(A.58) into the two market-share equations and subtracting gives

$$\begin{aligned} & \frac{1}{\sigma - 1} \left[\ln \frac{B_{2h}}{1 - B_{2h}} - \ln \frac{B_{2l}}{1 - B_{2l}} + \frac{2B_{2h} - 1}{B_{2h}(1 - B_{2h})} - \frac{2B_{2l} - 1}{B_{2l}(1 - B_{2l})} \right] \\ & \quad = \frac{\psi_2/P_h}{\sigma - 1} \left[\frac{1}{B_{2h}(1 - B_{2h})} + \frac{1}{B_{2l}(1 - B_{2l})} \right] - \ln \frac{\kappa q_{2l}}{P_l} - \left(1 - \frac{\kappa q_{2l}}{P_l} \right). \end{aligned}$$

The expression in square brackets on the left is strictly increasing in the market share, while the right-hand side is strictly positive whenever $\psi_2 > 0$. Hence

$$B_{2h} > B_{2l}. \tag{A.68}$$

Combining the market-share equations with firm 2's binding high-income IC and firm 1's slack IC then gives

$$\left(1-\frac{P_l}{P_h}\right)\left(\frac{p_{2l}-p_{1l}}{P_l}\right)=\frac{1}{\sigma-1}\left[\ln\frac{B_{2h}}{1-B_{2h}}-\ln\frac{B_{2l}}{1-B_{2l}}\right]+\text{IC slack of firm 1}>0, \tag{A.69}$$

$$\frac{p_{2h}-p_{1h}}{P_h}-\frac{P_l}{P_h}\left(\frac{p_{2l}-p_{1l}}{P_l}\right)=-\ln\frac{\kappa q_{2l}}{P_l}+\text{IC slack of firm 1}>0. \tag{A.70}$$

Therefore

$$\frac{p_{2h}-p_{1h}}{P_h}>0, \qquad \frac{p_{2l}-p_{1l}}{P_l}>0. \tag{A.71}$$

We also use one implication of profit maximization. Whenever a firm's high-income IC binds, it can withdraw its high-end product while leaving its low-end product unchanged. Since the high-income household was indifferent between the two products, this deviation leaves the firm's high-income customer base unchanged and replaces the high-end contribution margin by the low-end contribution margin. The latter is strictly positive by (A.58), so optimality implies

$$p_{jh}-\kappa q_{jh}\geq p_{jl}-\kappa q_{jl}>0. \tag{A.72}$$

Now suppose both high-income IC constraints bind. We first show that the same orderings continue to hold. Direct substitution of (A.55)–(A.58) gives

$$\frac{p_{2h}-p_{1h}}{P_h}=\frac{\frac{4}{\sigma-1}\left[2B_{2h}-1-\frac{\psi_2-\psi_1}{P_h}\right]}{1-(2B_{2h}-1)^2}, \tag{A.73}$$

$$\begin{aligned}\frac{p_{2l}-p_{1l}}{P_l}=\frac{2}{1-(2B_{2l}-1)^2}\Bigg\{&(2B_{2l}-1)\left[\frac{2}{\sigma-1}+\frac{P_h-P_l}{P_l}\frac{\psi_1+\psi_2}{P_h}\right]\\ &+\frac{\psi_2-\psi_1}{P_h}\left[\frac{2}{\sigma-1}-\frac{P_h-P_l}{P_l}\right]\Bigg\}.\end{aligned} \tag{A.74}$$

Subtracting the two binding IC constraints and using the two market-share equations yields

$$\left(1-\frac{P_l}{P_h}\right)\frac{p_{2l}-p_{1l}}{P_l}=\frac{1}{\sigma-1}\left[\ln\frac{B_{2h}}{1-B_{2h}}-\ln\frac{B_{2l}}{1-B_{2l}}\right]. \tag{A.75}$$

If the branch approaches a constrained symmetric limit, extend it to negative values of the signed brand gap by relabeling the two firms. Then P_l/P_h and $(\psi_1+\psi_2)/P_h$ are even functions of the signed gap, while $2B_{2h}-1$, $2B_{2l}-1$, and $(\psi_2-\psi_1)/P_h$ are odd. Differentiating the two market-share equations and the difference between the two binding IC constraints at the symmetric limit gives

$$\frac{d}{d\Delta}(B_{2h}-B_{2l})\Big|_{0^+}>0, \qquad \frac{d}{d\Delta}(\psi_2-\psi_1)\Big|_{0^+}>0, \qquad \frac{dB_{2h}}{d\Delta}\Big|_{0^+}>0. \tag{A.76}$$

For completeness, eliminating the three derivatives from the linearized system gives numerators

$$\frac{P_h-P_l}{P_l}\left[\frac{6}{\sigma-1}\frac{\kappa q_l}{P_l}+\left(\frac{P_h-P_l}{P_l}\right)^2\left(\frac{2\psi}{P_h}\right)^2+3\frac{P_h-P_l}{P_l}\frac{\kappa q_l}{P_l}\frac{2\psi}{P_h}+\frac{P_h-P_l}{P_l}\left(\frac{2\psi}{P_h}\right)^2\right],$$
$$\frac{P_h-P_l}{P_l}\left[\frac{4}{\sigma-1}\frac{\kappa q_l}{P_l}+\left(\frac{P_h-P_l}{P_l}\right)^2\left(\frac{2\psi}{P_h}\right)^2+\left(\frac{P_h-P_l}{P_l}\right)^2\frac{2\psi}{P_h}+2\frac{P_h-P_l}{P_l}\frac{\kappa q_l}{P_l}\frac{2\psi}{P_h}\right],$$

respectively, over a common strictly positive denominator. Hence all three signs in (A.76) are strict. If a both-binding segment begins instead when firm 1's previously slack IC becomes binding, the same inequalities hold at entry by continuity from (A.68)–(A.71).

These inequalities cannot reverse along a regular both-binding segment. Suppose first that $B_{2h} = B_{2l}$ at a first crossing. Equation (A.75) would then imply $p_{2l} = p_{1l}$. Substituting this equality into (A.56) and (A.58), the difference between the low-end quality distortions is strictly positive and equal to the difference in normalized low-end qualities. The difference between the two binding high-income IC constraints then requires

$$\frac{p_{2h}-p_{1h}}{P_h}=\ln\frac{q_{1l}}{q_{2l}}>\frac{\kappa(q_{1l}-q_{2l})}{P_l},$$

whereas (A.73) implies the reverse strict inequality. This is a contradiction. Hence $B_{2h} > B_{2l}$ throughout the both-binding segment.

Similarly, $\psi_2 - \psi_1$ cannot return to zero. If $\psi_2 = \psi_1$ while $B_{2h} > B_{2l}$, equation (A.75) implies $B_{2l} > 1/2$. Equations (A.56) and (A.58) then imply $q_{2l} > q_{1l}$ by an amount that is strictly smaller than the log quality difference required by the two binding IC constraints. This again contradicts (A.73). Thus $\psi_2 > \psi_1$, and (A.75) gives $p_{2l} > p_{1l}$.

Finally, (A.72) implies

$$\frac{\psi_2}{P_h}<B_{2h},\qquad \frac{\psi_1}{P_h}<1-B_{2h},\qquad \frac{\psi_1+\psi_2}{P_h}<1.$$

If $p_{2h} \le p_{1h}$, equation (A.73) would imply

$$\frac{\psi_2-\psi_1}{P_h}\ge 2B_{2h}-1.$$

Since $p_{2l} > p_{1l}$, the two binding IC constraints would then imply $q_{2l} > q_{1l}$, which by (A.56) requires

$$\frac{\psi_2-\psi_1}{P_h}<\frac{\psi_1+\psi_2}{P_h}(2B_{2l}-1)<2B_{2l}-1<2B_{2h}-1,$$

a contradiction. Hence, throughout every both-binding segment,

$$B_{2h}>B_{2l},\qquad \psi_2>\psi_1,\qquad p_{2h}>p_{1h},\qquad p_{2l}>p_{1l}. \tag{A.77}$$

These orderings also rule out a branch on which only firm 1's high-income IC binds. Firm 2 is the first firm to become constrained, and on a both-binding segment $\psi_2 > \psi_1$. Hence, if one multiplier returns to zero, it must be ψ_1.

Any allocation with active screening has inequality above income inequality. Suppose first that only firm 2's high-income IC binds. Since firm 1 is unconstrained, equations (A.57)–(A.58) and (A.77) imply $X > 1$ and $Y > 1$. Combining (A.63) and (A.65) gives

$$\begin{aligned}\ln C_h - \ln C_l - \ln\left(\frac{1+\alpha}{1-\alpha}\right) &= X - \ln X - Y + \ln Y \\ &\quad + \frac{1}{\sigma-1}\left[\ln\frac{1-B_{2l}}{1-B_{2h}} + \frac{1}{B_{2l}} - \frac{1}{B_{2h}}\right]. \end{aligned} \tag{A.78}$$

The last line is strictly positive because $B_{2h} > B_{2l}$. If $X \geq Y$, the first line is nonnegative because $v - \ln v$ is increasing for $v > 1$. If $Y > X$, the equilibrium equations imply that the positive term in the second line exceeds $Y - X$, while the derivative of $v - \ln v$ lies between zero and one on $[X, Y]$. Equation (A.78) is therefore strictly positive.

Now suppose both high-income IC constraints bind. The same substitutions give

$$\ln C_h - \ln C_l - \ln\left(\frac{1+\alpha}{1-\alpha}\right) = X - \ln X - Y + \ln Y \tag{A.79}$$

$$\begin{aligned} &- \ln\frac{\kappa q_{1l}}{P_l} - \left(1 - \frac{\kappa q_{1l}}{P_l}\right) \\ &+ \frac{1}{\sigma-1}\Bigg[\ln\frac{1-B_{2l}}{1-B_{2h}} + \frac{1}{B_{2l}} - \frac{1}{B_{2h}} \\ &\qquad + \frac{\psi_1}{P_h}\left(\frac{1}{B_{2h}(1-B_{2h})} + \frac{1}{B_{2l}(1-B_{2l})}\right)\Bigg]. \end{aligned} \tag{A.80}$$

The terms after the first line are strictly positive. If $Y \geq 1$, the preceding argument applies. If $Y < 1$, $p_{2l} > p_{1l}$ and the positive low-end margin imply

$$Y > \frac{p_{1l}}{P_l} > \frac{\kappa q_{1l}}{P_l}.$$

Therefore

$$-\ln\frac{\kappa q_{1l}}{P_l} - \left(1 - \frac{\kappa q_{1l}}{P_l}\right) > Y - \ln Y - 1.$$

Equation (A.72) implies $X > 1$, so $X - \ln X > 1$. Hence (A.80) is again strictly positive. We have shown that every allocation with active screening satisfies

$$\ln C_h - \ln C_l > \ln\left(\frac{1+\alpha}{1-\alpha}\right). \tag{A.81}$$

This proves the proposition whenever the equal-brand limit is unconstrained.

Constrained equal-brand limit. It remains to consider the case in which both high-income IC constraints are already binding in the equal-brand limit. Here it is useful to define one scalar that summarizes average low-end resource use,

$$z_0 \equiv \lim_{\delta\downarrow 0} \frac{\kappa}{P_h(\delta)}\left[B_{2l}(\delta)q_{2l}(\delta) + (1 - B_{2l}(\delta))q_{1l}(\delta)\right]. \tag{A.82}$$

At the symmetric limit, the two firms have identical qualities and prices and both high-income IC constraints bind. Taking the limit in (A.64) and (A.65), together with the firm first-order conditions, gives

$$\lim_{\delta\downarrow 0}\left[\ln C_h(\delta)-\ln C_l(\delta)\right]=-\ln z_0=\alpha\left(1+\frac{2}{\sigma-1}\right)(1+z_0). \tag{A.83}$$

Thus z_0 is uniquely determined by this scalar equation. Notice that this characterization uses only the welfare-relevant product z_0 and does not require interpreting the separate limiting values of P_h and P_l as marginal price indices at exact symmetry.

For an asymmetric allocation with active screening, define analogously

$$z\equiv\frac{\kappa}{P_h}\left[B_{2l}q_{2l}+(1-B_{2l})q_{1l}\right]. \tag{A.84}$$

A direct substitution of the firm first-order conditions into $X+(P_l/P_h)Y$ gives

$$X+\frac{P_l}{P_h}Y=\left(1+\frac{2}{\sigma-1}\right)(1+z)+\frac{2}{\sigma-1}F, \tag{A.85}$$

where

$$\begin{aligned}F\equiv\;&\frac{(2B_{2h}-1)\left[\left(2-\frac{\psi_1+\psi_2}{P_h}\right)(2B_{2h}-1)-\frac{\psi_2-\psi_1}{P_h}\right]}{1-(2B_{2h}-1)^2}\\&+\frac{P_l}{P_h}\frac{(2B_{2l}-1)\left[\left(2+\frac{\psi_1+\psi_2}{P_h}\right)(2B_{2l}-1)+\frac{\psi_2-\psi_1}{P_h}\right]}{1-(2B_{2l}-1)^2}.\end{aligned} \tag{A.86}$$

This is the only additional object we introduce in the constrained-limit comparison; it collects a long expression that appears repeatedly in the expenditure decomposition.

The orderings in (A.77) imply

$$D_{\mathrm{KL}}(B_{2l}\|B_{2h})+2F\geq 0,\qquad D_{\mathrm{KL}}(B_{2l}\|B_{2h})+2\alpha F\geq 0. \tag{A.87}$$

To prove this, the first term in (A.86) is positive. If the second is nonnegative, there is nothing to show. Otherwise $B_{2l}<1/2$. Write $v=1-2B_{2l}>0$ only for the next three displays. Negativity of the second term implies

$$\frac{\psi_2-\psi_1}{P_h}>\left(2+\frac{\psi_1+\psi_2}{P_h}\right)v,$$

and (A.77) then implies $2B_{2h}-1>2v$. Since $P_l/P_h<1$, the expression in (A.86) is bounded below by

$$-\frac{v(2B_{2h}-1-3v)}{1-v^2}.$$

If $v\geq(2B_{2h}-1)/3$, this lower bound is nonnegative. Otherwise Pinsker's inequality gives

$$D_{\mathrm{KL}}(B_{2l}\|B_{2h})\geq 2(B_{2h}-B_{2l})^2=\frac{(2B_{2h}-1+v)^2}{2},$$

and direct multiplication shows

$$\frac{(2B_{2h}-1+v)^2}{2} \geq \frac{2v(2B_{2h}-1-3v)}{1-v^2}.$$

This proves the first inequality in (A.87). The second follows immediately if $F \geq 0$; if $F < 0$, it follows from $0 < \alpha < 1$.

We can now complete the endpoint comparison. If $z < z_0$, equation (A.65), Jensen's inequality, and (A.77) imply

$$\ln C_h - \ln C_l > -\ln z > -\ln z_0 = \lim_{\delta \downarrow 0} \left[\ln C_h(\delta) - \ln C_l(\delta)\right].$$

If instead $z \geq z_0$, combine (A.64), (A.83), (A.85), and (A.87). The two high-income IC slacks are nonnegative, so

$$\begin{aligned}
&\ln C_h - \ln C_l - \lim_{\delta \downarrow 0} \left[\ln C_h(\delta) - \ln C_l(\delta)\right] \\
&\quad \geq \alpha \left(1 + \frac{2}{\sigma - 1}\right)(z - z_0) + \frac{1}{\sigma - 1}\left[D_{\mathrm{KL}}(B_{2l} \| B_{2h}) + 2\alpha F\right] \geq 0.
\end{aligned}$$

The inequality is strict for every $\Delta > 0$ on an active branch.

Combining the unconstrained and constrained equal-brand cases proves the proposition. Equality can occur only while screening remains inactive. ■

B Quantitative Model Appendix

B.1 Computational Algorithm

This appendix describes the algorithm we use to solve and calibrate the multi-sector model of Section 5. Sectors interact only through a small set of aggregate objects—the marginal price indices P_l and P_h and aggregate profits—while within each sector the firms' optimal choices are available in closed form given those aggregates (Proposition 5). We exploit this structure and solve all 897 sectors (roughly 6.2 million firms) simultaneously by iterating on the aggregate objects, evaluating every firm-level condition in closed form at each step. The sector-specific parameters $\{\chi_s, \omega_s\}$ are calibrated inside the same fixed point, so that calibration and equilibrium computation are performed in a single loop and the model hits its targets exactly at the solution.

Normalizations. We normalize the wage to one and measure quality in units of labor per unit mass of each household type (each type has mass one-half). With this convention, the perfectly-competitive qualities are $q_\tau^{PC} = L_\tau/2$ and the labor-market clearing condition (2.10) reads

$$\sum_s \omega_s \sum_j (B_{jl}q_{jl} + B_{jh}q_{jh}) = 1.$$

Because utility is logarithmic in quality, this choice of units shifts log consumption of both household types by a common constant and cancels from all reported welfare and inequality comparisons.

Brand values. Rather than simulating draws from the sectoral Pareto distributions, we assign each firm the expected value of its order statistic. In a sector of type s, the firm with rank k among N_s i.i.d. draws from a Pareto distribution with shape χ_s and scale one receives

$$\eta_{(k)} = \mathbb{E}\Big[\eta_{(k:N_s)}\Big] = \frac{B(N_s - k + 1,\, k - 1/\chi_s)}{B(N_s - k + 1,\, k)}, \tag{B.1}$$

where $B(\cdot,\cdot)$ is the Euler beta function. We evaluate (B.1) as the exponential of differences of log-beta functions, which is numerically stable even in sectors with hundreds of thousands of firms. This construction makes the brand-value profile a deterministic and smooth function of (N_s, χ_s), eliminating simulation error and allowing the calibration of χ_s described below to proceed by fixed-point iteration.

B.2 Algorithm for the Calibrated Equilibrium

Let $S_{j\tau} \equiv e^{(\sigma-1)(\ln q_{j\tau} - p_{j\tau}/P_\tau + \eta_j)}$ denote firm j's demand kernel in segment $\tau \in \{l, h\}$, so that market shares are $B_{j\tau} = S_{j\tau}/\sum_{j'} S_{j'\tau}$, with the sum taken within the firm's sector. The algorithm iterates jointly on the firm-level objects $\{S_{jl}, S_{jh}, \psi_j\}$, where ψ_j is the multiplier on the high-income IC constraint, the aggregates P_h and $\widetilde{P} \equiv P_h/P_l$, and the calibrated parameters $\{\chi_s\}$. It consists of the following steps.

0) Start with a guess for $\{S_{jl}, S_{jh}, \psi_j\}$, P_h, $\widetilde{P}$, and $\{\chi_s\}$. We initialize $P_h = 1$, $\widetilde{P} = \frac{1+\alpha}{1-\alpha}$, $\psi_j = 0$, a common χ_s across sectors, and the demand kernels implied by symmetric qualities.

1) Given the guess, compute every firm's market shares, qualities, and prices from the optimality conditions of Proposition 5, handling the binding and slack IC regimes by complementary slackness. Obtain updated firm-level objects $\{S_{jl}, S_{jh}, \psi_j\}$.

2) Aggregate across firms and sectors to obtain updated values of P_h, $\widetilde{P}$, and the calibrated parameters $\{\omega_s, \chi_s\}$. If the maximal distance between the guess and the update across all blocks is below 10^{-5}, stop; otherwise update the guess by a damped convex combination and return to step (1).

Step 1 – Firms' optimal choices. Market shares follow from normalizing the demand kernels within each sector. Qualities are given by equations (3.11) and (3.12), and prices by (3.14) together with the binding IC constraint, $p_{jh} = p_{jl} + P_h (\ln q_{jh} - \ln q_{jl})$. To determine the set of firms for which the IC constraint binds, we invert the high-income pricing condition (3.13) to recover the implied multiplier

$$\psi'_j = B_{jh} P_h - (p_{jh} - P_h)(\sigma - 1) B_{jh} (1 - B_{jh}). \tag{B.2}$$

Firms with $\psi'_j \leq 0$ are in the slack regime: we set $\psi'_j = 0$ and replace their high-income price with the unconstrained oligopoly price (3.3). This regime assignment implements the Kuhn–Tucker conditions exactly at the fixed point. Finally, the updated demand kernels $S'_{j\tau}$ are computed from the resulting qualities and prices.

Step 2 – Aggregation and calibration. Four updates are performed. First, the sector weights ω_s are set so that model sectoral sales shares match their empirical counterparts exactly (a closed-form rescaling, given firm sales from step 1). Second, the Pareto shapes are updated multiplicatively toward the concentration target,

$$\chi'_s = \frac{T4_s^{\text{model}}}{T4_s^{\text{data}}} \, \chi_s, \tag{B.3}$$

where $T4_s$ is the sales share of the four largest firms in sector s. The update exploits that concentration is decreasing in χ_s: a lower shape parameter generates fatter-tailed brand values and a higher top-4 share. Third, P_h is updated to impose labor-market clearing, $P'_h = P_h / \sum_s \omega_s \sum_j (B_{jl} q_{jl} + B_{jh} q_{jh})$. Fourth, $\widetilde{P}$ is updated to impose the household budget constraints, under which relative spending of the two types equals relative income: $\widetilde{P}' = \frac{1+\alpha}{1-\alpha} / \frac{\sum_s \omega_s \sum_j p_{jh} B_{jh}}{\sum_s \omega_s \sum_j p_{jl} B_{jl}} \cdot \widetilde{P}$.

Convergence is declared when the largest absolute residual across all blocks—demand kernels, multipliers, price indices, and concentration targets—falls below 10^{-5}. We use a damping factor of 0.25 on the fast block $(S_{jl}, S_{jh}, \psi_j, P_h)$ and 0.0125 on the slow block $(\widetilde{P}, \chi_s)$, whose elements interact through the within-sector optimality conditions; ψ_j is projected onto $\psi_j \geq 0$ after each update. From a cold start the algorithm converges in roughly 14,000 iterations, a few minutes on a standard desktop.

B.3 Counterfactual Allocations

Perfectly competitive allocation. Given the calibrated $\{\chi_s, \omega_s\}$ and the implied brand values, the perfectly-competitive allocation is available in closed form (Proposition 2): qualities are $q_\tau^{PC} = L_\tau/2$ and market shares are proportional to $e^{(\sigma-1)\eta_j}$ within each sector. Welfare for each type is evaluated using equation (2.12), aggregated across sectors with the weights ω_s.

Island economy. In the island economy, qualities equal their perfectly-competitive levels (Proposition 3) and market shares are identical across segments, so only the fixed point in market shares remains. We iterate on the high-income demand kernel alone: given S_{jh}, compute B_{jh}, set prices to the unconstrained oligopoly price (3.3), and update the kernel; the loop uses the same damping and tolerance as the main algorithm and converges quickly since $\{\chi_s, \omega_s\}$ are held at their calibrated values and P_h is also held fix as it's equal to the perfectly-competitive allocation P_h.

Decomposition. The decomposition in Table 4 is computed by evaluating the welfare formula (2.12) at counterfactual combinations of the equilibrium objects: the misallocation margin evaluates welfare at the perfectly-competitive qualities but the market-equilibrium shares $B_{j\tau}$, and the quality margin is the remainder of the total gap.

B.4 The 1992 Economy and the HHI Variant

For the 1992 recalibration, we hold the sector weights ω_s and all common parameters at their 2022 values and re-run the algorithm of Section B.2 with the χ_s update retargeted to the 1992 top-4 sales shares; sectoral sales shares are then an equilibrium outcome rather than a target. We warm-start the iteration from the 2022 equilibrium, which leaves the fixed point unchanged but substantially reduces the number of iterations. The island-economy and perfectly-competitive allocations are recomputed for the 1992 parameters as described above.

The HHI-targeted calibration of Appendix B.5 uses the identical algorithm with the concentration update replaced by $\chi_s' = \left(\mathrm{HHI}_s^{\mathrm{model}}/\mathrm{HHI}_s^{\mathrm{data}}\right)\chi_s$. Mirroring the Census definition, the model HHI is computed from the squared market shares of the 50 largest firms in each sector.

B.5 Alternative Calibration: Sectoral HHI

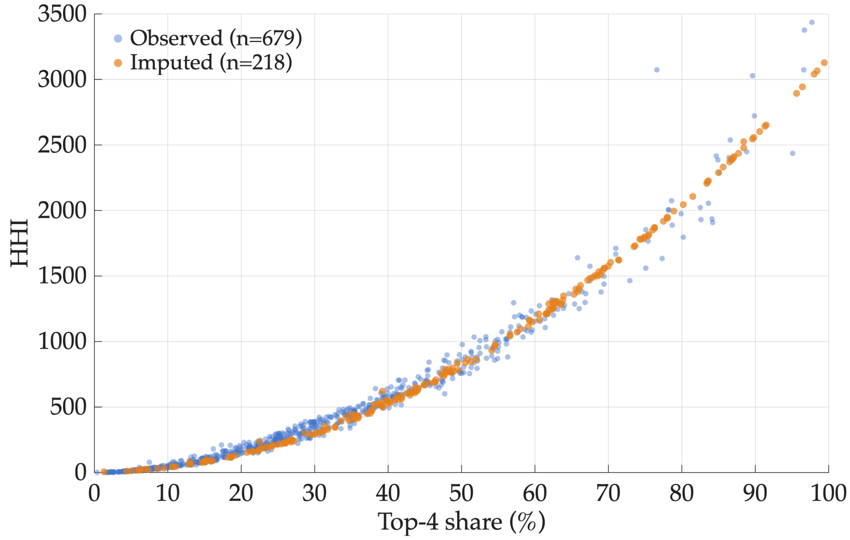
In this section, we present the results of the quantitative model if one were to target sectoral HHI instead of the top 4 sales share in each sector. Other than this change, the model as well as the calibration targets are identical to Section 5.

In the Establishment and Firm Size Statistics data tables from the 2022 Economic Census, the HHI is not reported for 220 sectors. For these sectors, we impute the HHI using the other 2 available moments: the share of sales accounted for by the top 4 and top 20 firms. Figure B.1 visualizes the imputation. Specifically, we estimate by OLS, on the 678 sectors with an observed HHI, the regression

$$\mathrm{HHI}_j \;=\; \alpha \;+\; \beta_1\,\mathrm{top4}_j^2 \;+\; \beta_2\left(\mathrm{top20}_j - \mathrm{top4}_j\right)^2 \;+\; \varepsilon_j,$$

where the sales shares are expressed in percent and the HHI is on its standard 0–10,000 scale. The specification follows from the definition of the HHI as the sum of squared market shares. If the top 4 firms split their combined share equally, they contribute $\text{top4}^2/4$ to the index; if firms 5 through 20 split theirs equally, they contribute $(\text{top20} - \text{top4})^2/16$. The estimated coefficients, $\hat{\alpha} = 4.5$, $\hat{\beta}_1 = 0.316$, and $\hat{\beta}_2 = 0.050$, are close to these equal-split benchmarks of 0.25 and 0.0625, with $\hat{\beta}_1$ exceeding its benchmark because sales are concentrated within the top 4 firms. The regression yields an R^2 of 0.98, and we use its fitted values to impute the HHI in the censored sectors.[17]

Figure B.1: Imputation of Missing Sectoral HHIs



Notes: This figure plots the HHI of each 6-digit NAICS industry against the sales share of its largest 4 firms. Blue circles are industries with a reported HHI in the 2022 Economic Census; orange circles are the 218 industries with missing HHI, for which we impute the HHI from the top-4 and top-20 sales shares.

The two aggregate parameters, σ and α, as well as the number of firms per sector, N_s, remain the same as in the baseline calibration. Panel B of Table B.1 below reports the mean, median, and standard deviation of the estimated Pareto shape parameters.

Relative to the baseline calibration, targeting HHIs yields slightly higher estimates of the Pareto shape (5.5 vs 5.3). That is, there is less dispersion in brand value. Consequently, welfare losses due to the misallocation margin are smaller. However, our main quantitative results are unchanged: (i) cannibalization concerns within the firm lead to large increases in consumption inequality over income inequality (6.9 log points), and (ii) the majority of the distortions relative to the perfectly competitive allocation stem from product design, with a smaller role for the more standard channel of misallocation across firms.

[17] Three of the 220 censored sectors report only one of the two sales shares. For these sectors, we impute the HHI using the corresponding single-predictor regression: $\text{HHI}_j = \alpha + \beta\, \text{top4}_j^2 + \varepsilon_j$ ($\hat{\beta} = 0.319$, $R^2 = 0.977$) or $\text{HHI}_j = \alpha + \beta\, \text{top20}_j^2 + \varepsilon_j$ ($\hat{\beta} = 0.159$, $R^2 = 0.752$).

Table B.1: Calibrated Parameters

Panel A. Common parameters				
Parameter	Description	Value	Moment	Source
σ	Dispersion of taste shocks	3	Elasticity of substitution across firms	Literature
α	Spending inequality	0.31	Non-housing exp. inequality	PSID (2022)

Panel B. Sector-specific parameters						
Parameter	Description	Mean	Median	SD	Moment	Source
N_s	Number of firms	6,941	1,189	18,525	# firms in 6-digit NAICS	Census (2022)
χ_s	Pareto shape	5.5	5.5	0.9	HHI	Census (2022)
ω_s	Sector weight	0.001	0.0003	0.003	Sales share	Census (2022)

Notes: Mean, median, and standard deviation are computed across the 899 6-digit NAICS industries for sector-specific parameters. α is chosen to match the relative expenditures of households above and below median income, trimming the top and bottom 10% of households.

Table B.2: Aggregate Consumption Relative to the Perfectly-Competitive Allocation

	High-income	Low-income	Inequality
Market equilibrium	+1.9	−5.0	6.9
Quality margin	+2.2	−4.3	6.5
Misallocation margin	−0.3	−0.7	0.4

Notes: This table reports the log point difference in aggregate consumption of each income type relative to the perfectly-competitive allocation, and decomposes it into quality and misallocation margins. The first row corresponds to the market allocation. The quality margin isolates the effect of quality distortions, while the misallocation margin isolates the effect of distorted market shares.